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Interface-Aware Recommender Systems

Recommender Systems; Carouse interfaces;
User Modeling; Browsing Behavior

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Kempelen Institute of
Intelligent Technologies

Objectives of the Tutorial

- Introduce the important role interface design plays on user browsing/interaction behavior and how this impacts the recommendation feedback loop and user experience
- Overview recent developments, from user studies to computational works, along with challenges & directions for future research
- Provide tools, methods, and datasets for developing interface-aware systems and realistic user models

Outline of Tutorial

1. Overview of Interfaces and interaction paradigms [20 mins]
2. Interface-aware recommenders through the lens of carousels: challenges, open problems, and future work [45 mins]
 - a. Understanding how users browse
 - b. Building interface-aware user models
 - c. Interface-aware evaluation
3. Tools/resources for Interface-Aware Recommender Systems [10 mins]
 - a. CARE: An Infrastructure for Evaluation of Carousel-Based Recommender Interfaces
 - b. RecGaze Dataset: First eye tracking and interaction dataset in carousel recommenders
4. Discussion, Q&A [15 mins]

Slides / Resources



<https://github.com/santideleon/Tutorial-on-Interface-Aware-Recommender-Systems>



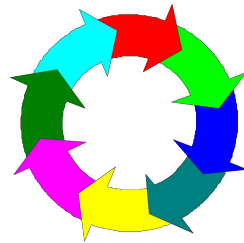
Overview of Interfaces and interaction paradigms

Anatomy of Traditional Recommender Systems

Input



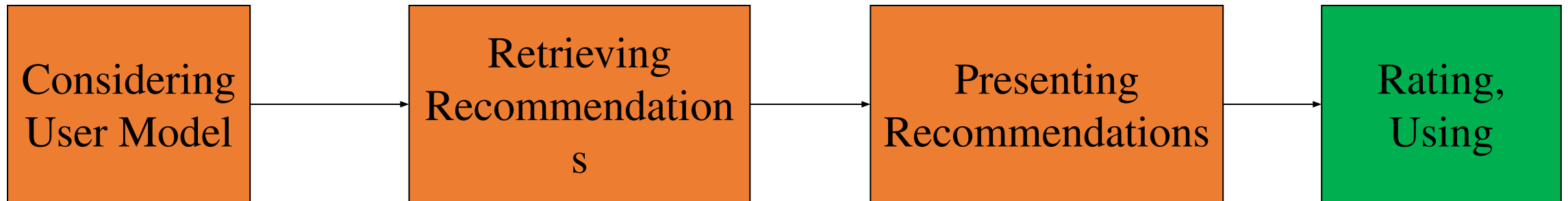
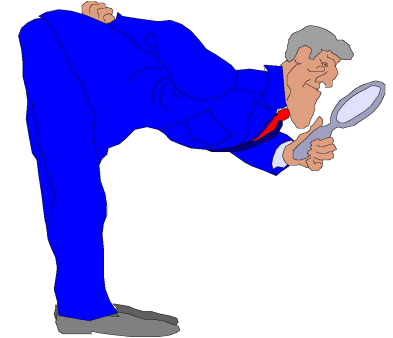
Process



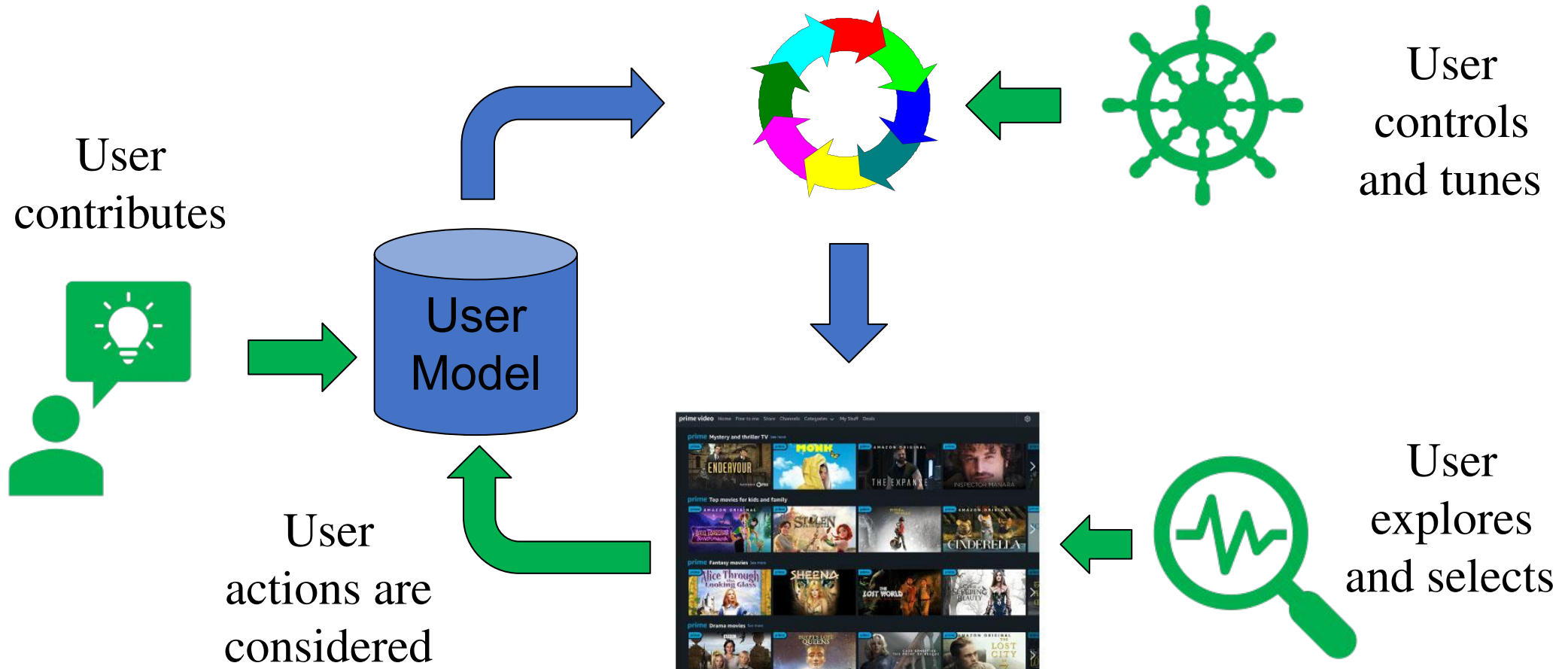
Result



Feedback



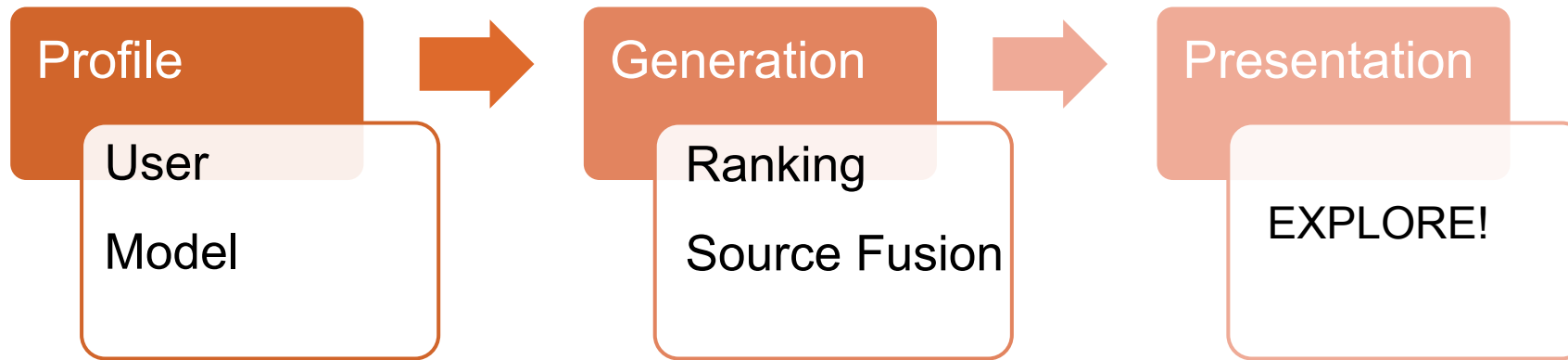
What is the point of having a interface?



Two groups of problems addressed by this tutorial

- Part 1: How to build interfaces that offer user more freedom to be more active and contribute to the process
 - What and where the users can contribute to the recommendation process?
 - What are the best practices in building user interfaces for RecSys?
- Part 2: How to improve recommender systems so they can leverage user actions in the interface, i.e., become “interface aware”
 - Understanding browsing behavior and its impact
 - Computational modeling of browsing behavior (realistic user models)
 - Designing evaluation metrics that are inline with interface layouts and browsing patterns

Three Places to Offer User Interface



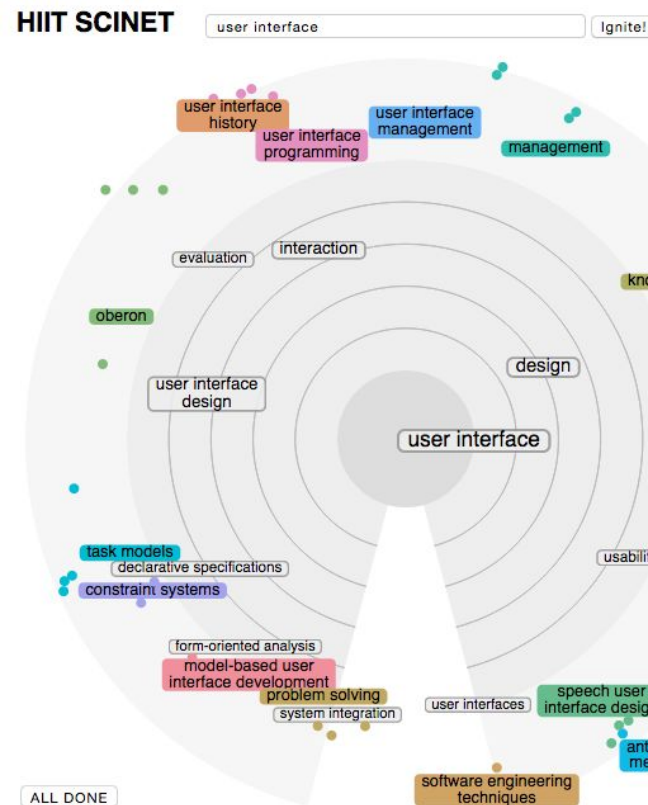
Interfaces to Refine User Profile

- Content-Based Filtering
 - Unigram (keyword) profiles
 - Semantic entities (artists, concepts, products)
- Collaborative Filtering
 - Peer selection
- Social Filtering
 - Trust in your friends

Your News: Keyword-Level Open User Profile

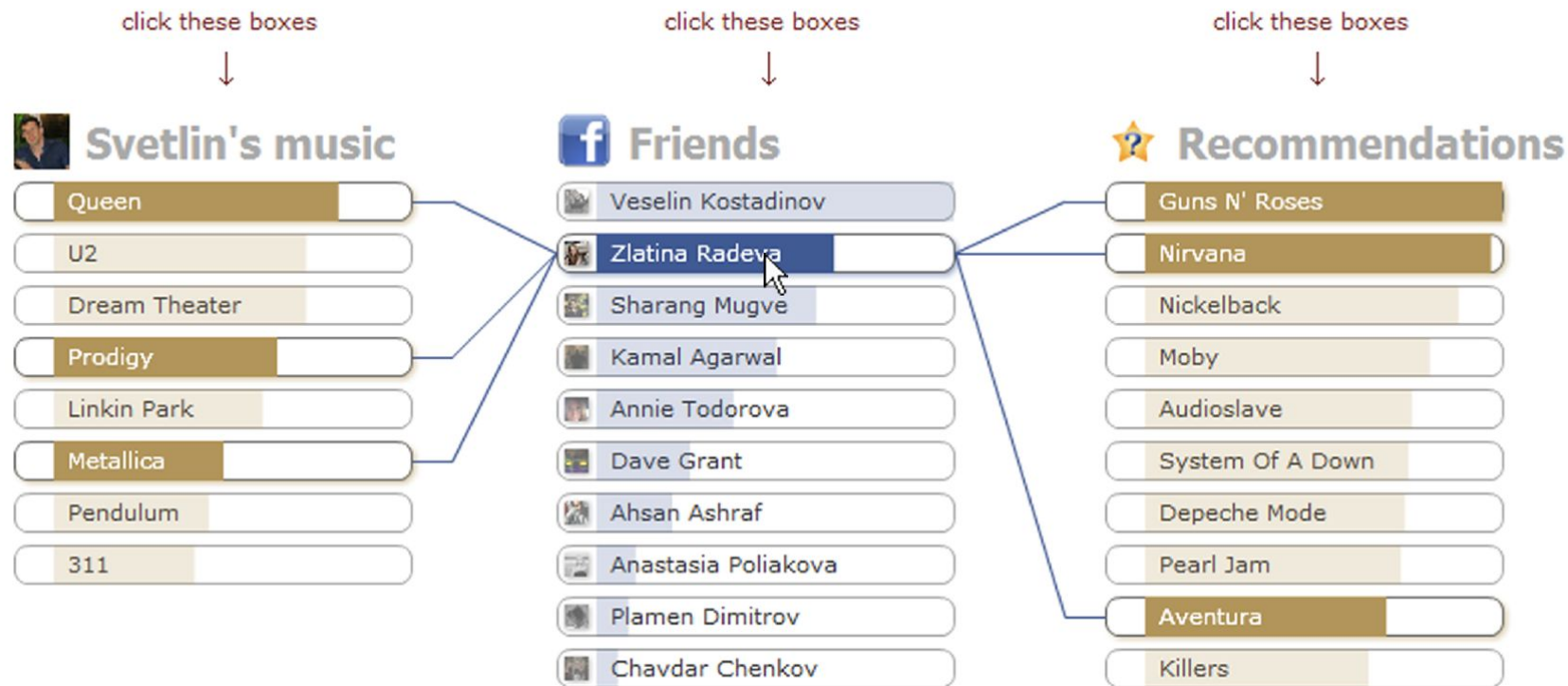


SciNet: Concept-Level Open User Profile

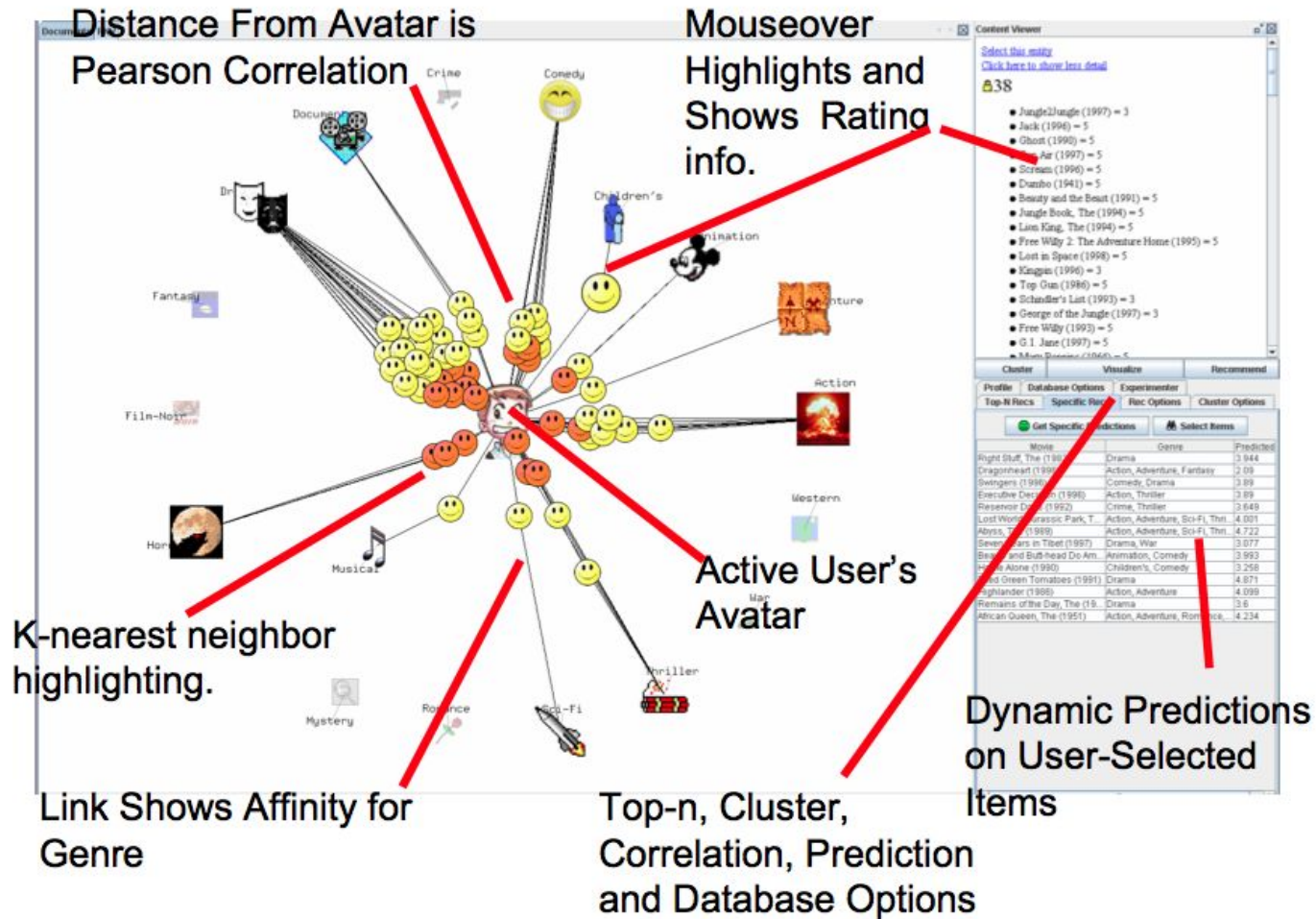


- ☐ **Abstract Specification of User Interfaces**
Ole Lauridsen (Proceedings of ACM CHI'95 Conference on Human Factors in Computing Systems, 1995-01-01)
user interface user interface design evaluation design
The paper discusses automation of user i...
- ☐ **Formalizing User Interface Requirements**
Kevin Schlueter, Marilyn Mantei (Proceedings of ACM INTERCHI'93 Conference on Human Factors in Computing Systems -- Adjunct Proceedings, 1993-01-01)
user interface design
User interface deficiencies often occur ...
- ☒ **Permissive user interfaces**
H Thimbleby (INTERNATIONAL JOURNAL OF HUMAN-COMPUTER STUDIES, 2001-01-01)
permissiveness user interface design design principles usability user interface programming
user interface design
User interfaces often only support one w...
- ☒ **User interfaces supporting the software process**
H U Kobialka, C Lewerentz (SOFTWARE PROCESS TECHNOLOGY, 1998-01-01)
user interface
This paper describes how user interfaces...
- ☐ **User Interface Tools**
Brad A. Myers, Dan R. Olsen (Proceedings of ACM CHI'94 Conference on Human Factors in Computing Systems, 1994-01-01)
user interface user interface design design
A user interface tool is any software th...
- ☐ **Ant colony metaphor applied in user interface recommendation**
J Sobecki (NEW GENERATION COMPUTING, 2008-01-01)
user interfaces recommendation ontologies ant colony metaphor user interface
user interface design user modelling metaphors design
In this paper web-based system user inte...
- ☐ **Appropriateness of user interfaces to tasks**
Sandrine Balbo, Dirk Draheim, Christof Lutteroth, Gerald Weber (International Workshop

Content-Based + Social Filtering: TasteWeights



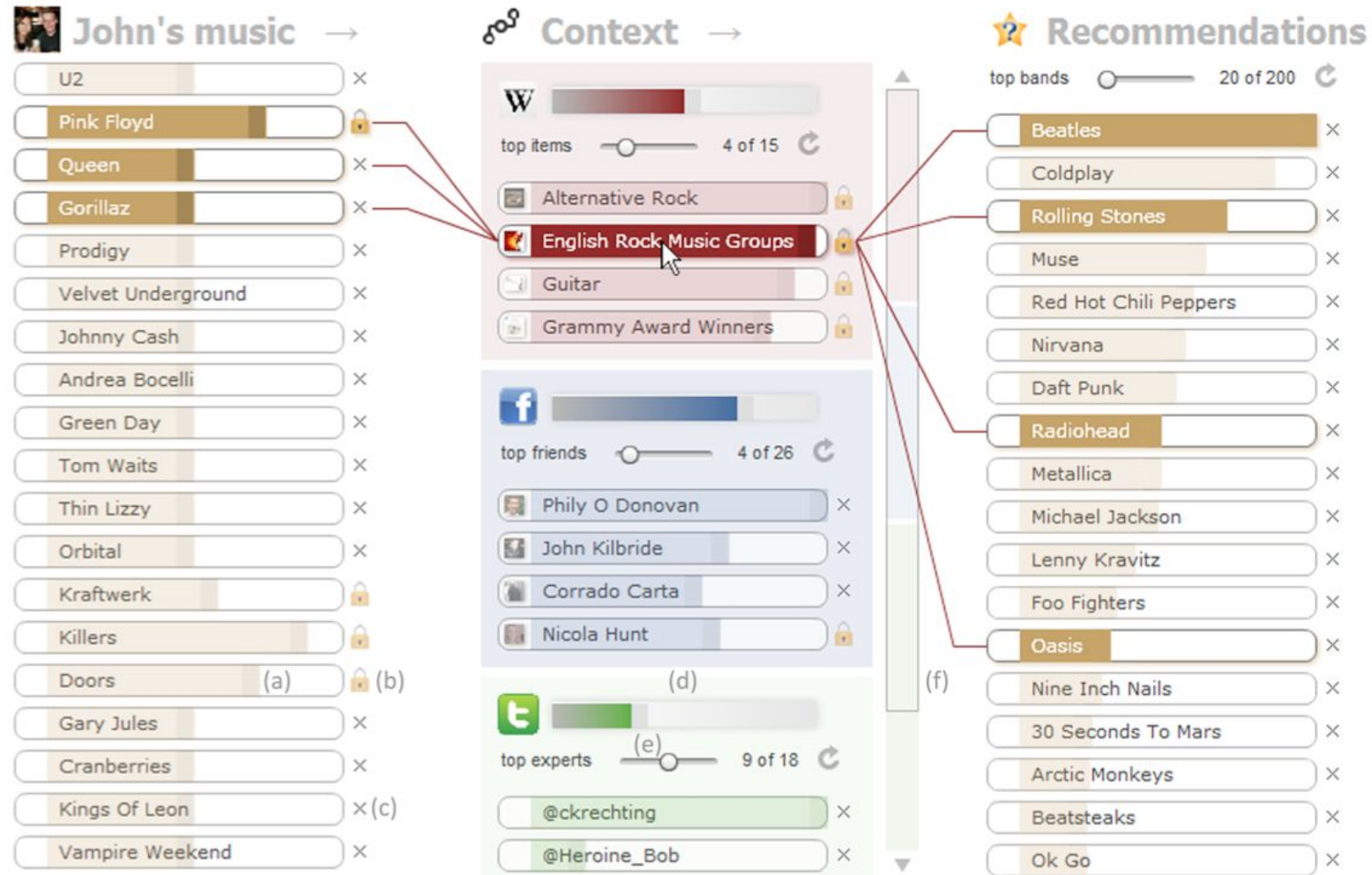
Collaborative Filtering: PeerChooser



Interfaces to Impact Relevance Assessment

- Choose the Recommendation Algorithm / Engine for the current need
- Tune your engine to the current context
 - Tag-based Parameter Tuning
 - Slider-Based Parameter Tuning
- Choose fusion weights for hybrid recommendation

TasteWeights: Choosing the Engine



Compound Critiques - Early GUI Interfaces

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>> Digital Cameras**Unit Critiques**

Shop for: Digital Cameras Computers Holidays



Product Found: Canon EOS 30

6.3 Megapixel CMOS sensor
7-point wide-area AF
High-performance DIGIC processor
100-1600 ISO speed range
Compatible with all Canon EF lenses and EX Speedlites
PictBridge, Canon Direct Print and Bubble Jet Direct compatible – no PC required

I've found the Camera I want! 

No lets start again 

Adjust your preferences to find the right camera for you

Manufacturer	X	Canon	X
Optical Zoom	↓	7x	↑
Memory (MB)	↓	512	↑
Weight (Grams)	↓	780	↑
Resolution	↓	6.2 M Pixels	↑
Size	X	Large	X
Case	X	Magnesium	X
Price	↓	995	↑

We have more matching cameras with the following:

1. Less Memory and Lower Resolution and Cheaper	EXPLAIN	PICK
2. Different Manufacturer and Less Zoom and Lighter	EXPLAIN	PICK
3. Lighter and Smaller and Different Case	EXPLAIN	PICK

Explain:

1. Less Memory and Lower Resolution and Cheaper

This Critique covers **153** other Digital Cameras

Less Memory
Current Value: 512 MB
Critique: Less Than
Remaining: (0 to 256 MB)

Lower Resolution
Current Value: 6.2 M Pixels
Critique: Less Than
Remaining: (1.4 to 5.9 M Pixels)

Cheaper
Current Value: 995 €
Critique: Less Than
Remaining: (75€ to 960€)

PICK

Movie Tuner: Tag-based Tuning of CF Engine

Movie Tuner (?)
Find similar movies with less or more of particular qualities. The movie list below will update as you indicate your preferences. Use the lock button to select multiple qualities.

This movie	What I want		
	less	ok	more
action	☐	☒	☐
violent	☒	☐	☐
dark	☐	☒	☐
crime	☐	☒	☐
funny	☐	☒	☐
Enter selection...	☐	☐	☐

lock

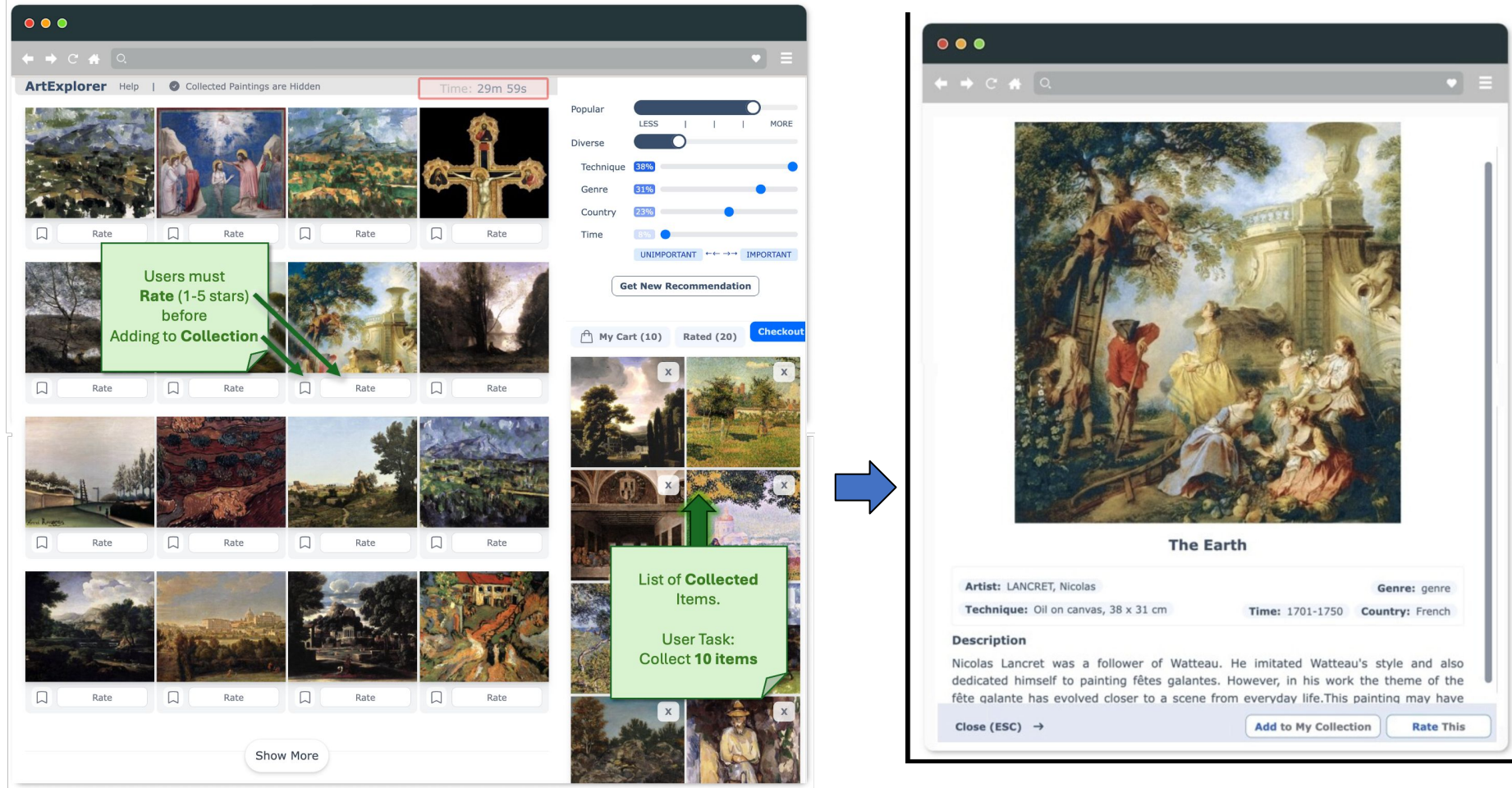
reset all

Similar to *Reservoir Dogs*, but **less violent**

- House of Games (1987)
- Out of Sight (1998)
- Killing, The (1956)
- Usual Suspects, The (1995)
- Four Rooms (1995)
- Blood Simple (1984)
- Get Shorty (1995)

Movie Tuner interface for *Reservoir Dogs* after a user applies the critique “less violent”.

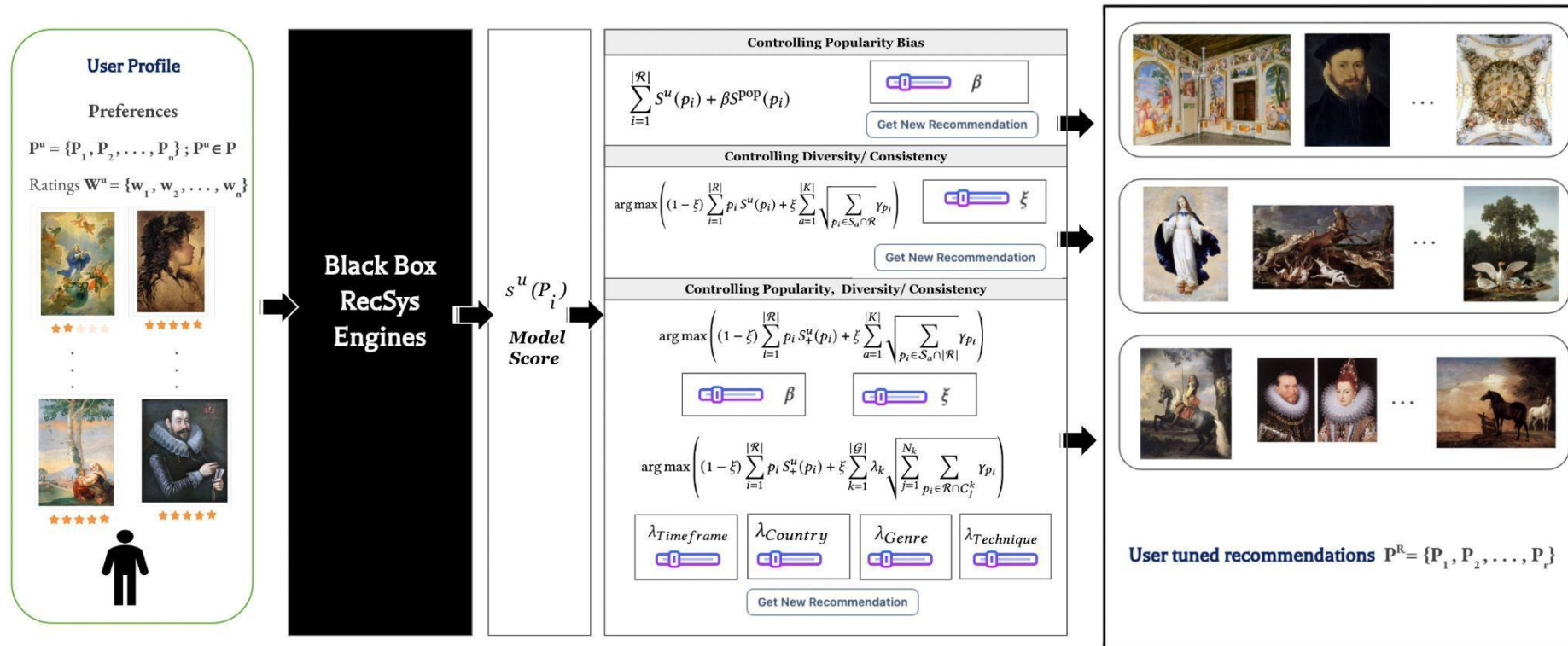
ArtEx: User Control over Diversity and Popularity



Hendrawan, R. A., Brusilovsky, P., Yilma, B. A., and Leiva, L. A. (2025) Recommending Paintings in Web Art Gallery with Adjustable Popularity and Diversity.

Workshop on Personalized Access to Cultural Heritage at the 33rd ACM Conference on User Modeling, Adaptation and Personalization, June 19, 2025..

ArtEx Engine: Learning from User Control



RelevanceTuner: Fusion Control in a Hybrid Recommender

Publication Similarity ⓘ		Topic Similarity ⓘ		Co-Authorship Similarity ⓘ		CN3 Interest Similarity ⓘ		Geographic Distance ⓘ	
5		5		5		5		5	
Profile	Relevance ▾		Name	Follow	Connect	Affiliation	Position		
	10 10 10 10 10		Chun-Hua Tsai			University of Pittsburgh	PhD Candidate		
	9 7 10 10 10	ⓘ	Peter Brusilovsky	Following X	Already contact X	University of Pittsburgh			
	7 5 10 4 10	ⓘ	Casey Dugan	Follow	Add as connection	IBM Research	Researcher		
	5 10 9 10	ⓘ	Jordan Ariel Barria-Pineda	Following X	Add as connection	University of Pittsburgh			
	10 10 10 5	ⓘ	Denis Parra	Follow	Add as connection	PUC Chile	Assistant Professor		

Presentation Interfaces: Helping User To Choose

- Conversational recommendation – No choice
 - Compare with Google old “I feel lucky”
- Ranked List – item rank helps to choose
- Clustering and grouping – clusters help to choose
- Spatial visualization – item location helps to choose
 - Relevance-based
 - Similarity-based
- Topological visualization – connection of items with other entities help to choose

Clustering Instead of Ranking

The top candidate according to your preferences

Manufacturer	Price	MegaPixels	Optical zoom	Memory type	Flash memory	LCD screen size	Depth	Weight	
Canon	\$242.00	5.0 MP	3x	CompactFlash Card	32 MB	1.8 in	1.37 in	8.3 oz	choose

We have more products with the following they are cheaper and lighter, but have fewer megapixels

Nikon	\$167.95	4 MP	3x	SD Memory Card	14 MB	1.8 in	1.4 in	4.6 oz	choose
Canon	\$230.00	4.1 MP	3x	CompactFlash Card	32 MB	1.5 in	1.09 in	6.53 oz	choose
Canon	\$180.00	3.3 MP	3x	SD Memory Card	16 MB	2 in	0.83 in	4.06 oz	choose
Canon	\$219.18	4.2 MP	4x	MultiMedia Card	16 MB	1.8 in	1.51 in	6.35 oz	choose
Canon	\$163.50	3.2 MP	4x	MultiMedia Card	16 MB	1.8 in	1.5 in	6.3 oz	choose
Canon	\$199.40	3.2 MP	2.2x	SD Memory Card	16 MB	1.5 in	1.4 in	5.8 oz	choose

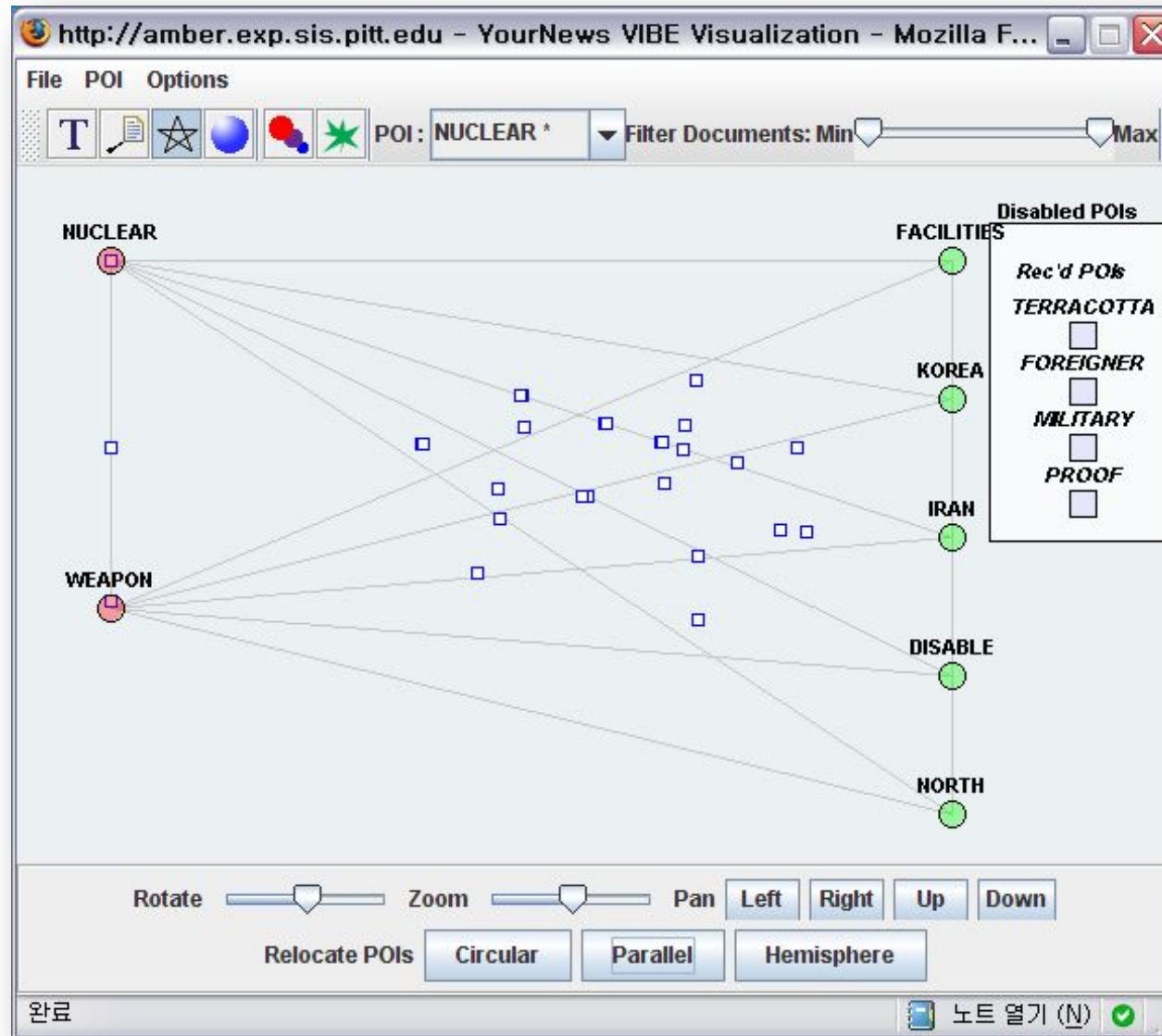
they have more megapixels and bigger screens, but are more expensive

Sony	\$365.00	7.2 MP	3x	Internal Memory	32 MB	2.5 in	1.5 in	6.9 oz	choose
Canon	\$439.99	7.1 MP	3x	SD Memory Card	32 MB	2 in	1.04 in	6 oz	choose
Fuji	\$253.00	6.3 MP	4x	XD-Picture Card	16 MB	2 in	1.4 in	7.1 oz	choose
Sony	\$336.00	7.2 MP	3x	Internal Memory	32 MB	2 in	1 in	5 oz	choose
Nikon	\$304.18	7.1 MP	3x	Internal Memory	13.5 MB	2 in	1.4 in	5.3 oz	choose
Olympus	\$334.00	7.4 MP	5x	XD-Picture Card	32 MB	2.0 in	1.7 in	7.1 oz	choose

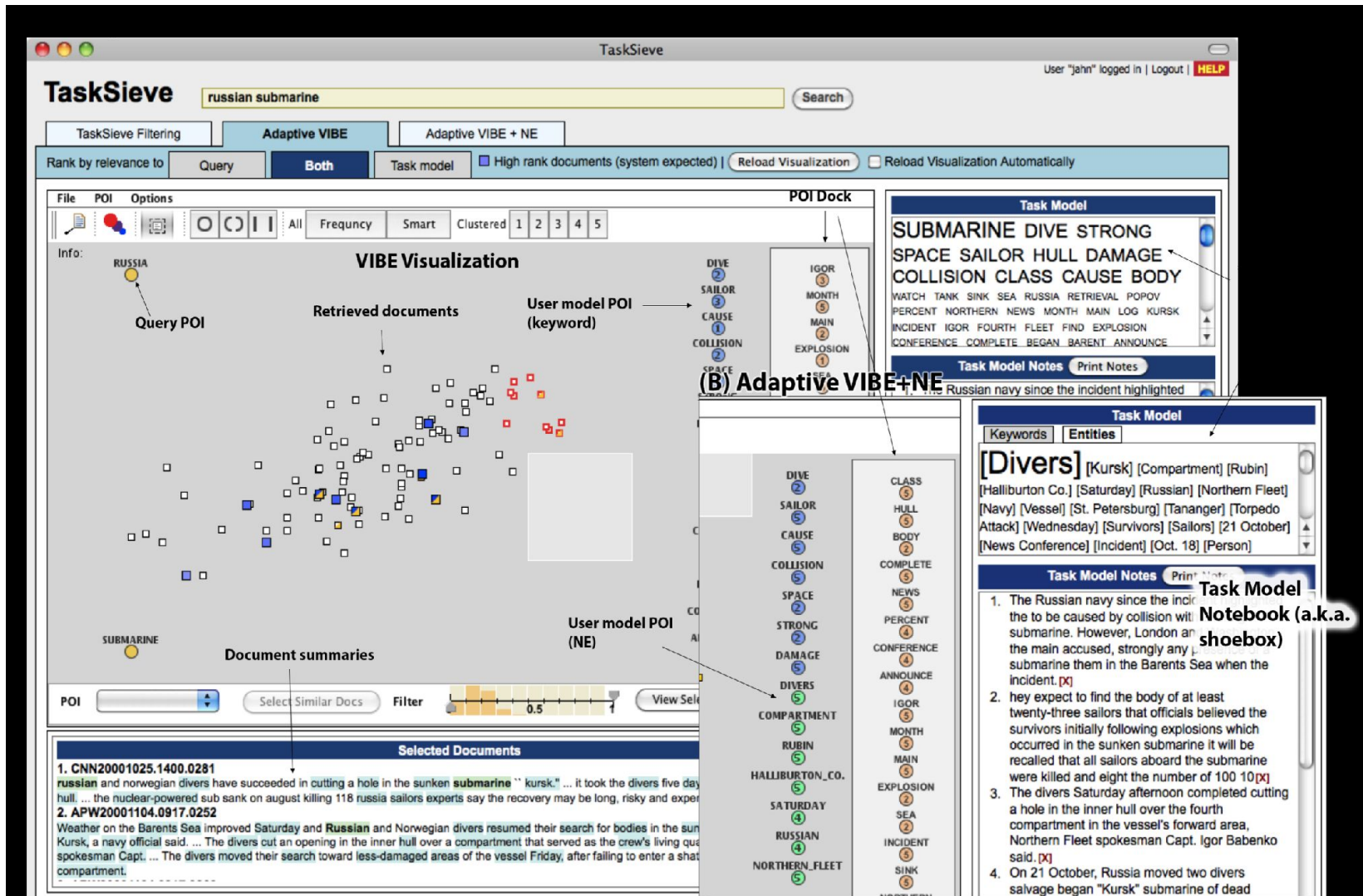
they are lighter and thinner, but have less flash memory

Pentax	\$238.99	5.3 MP	3x	Internal Memory	10 MB	1.8 in	0.8 in	3.7 oz	choose
Canon	\$273.18	4.0 MP	3x	SD Memory Card	16 MB	2 in	0.82 in	4.59 oz	choose
Nikon	\$329.95	5.1 MP	3x	Internal Memory	12 MB	2.5 in	0.8 in	4.2 oz	choose
Canon	\$316.18	5.3 MP	3x	SD Memory Card	16 MB	2 in	0.81 in	4.59 oz	choose
Casio	\$386.00	7.2 MP	3x	Internal Memory	8.3 MB	2.5 in	0.88 in	4.48 oz	choose
Fuji	\$309.18	6.3 MP	3x	XD-Picture Card	16 MB	2.5 in	1.1 in	5.5 oz	choose

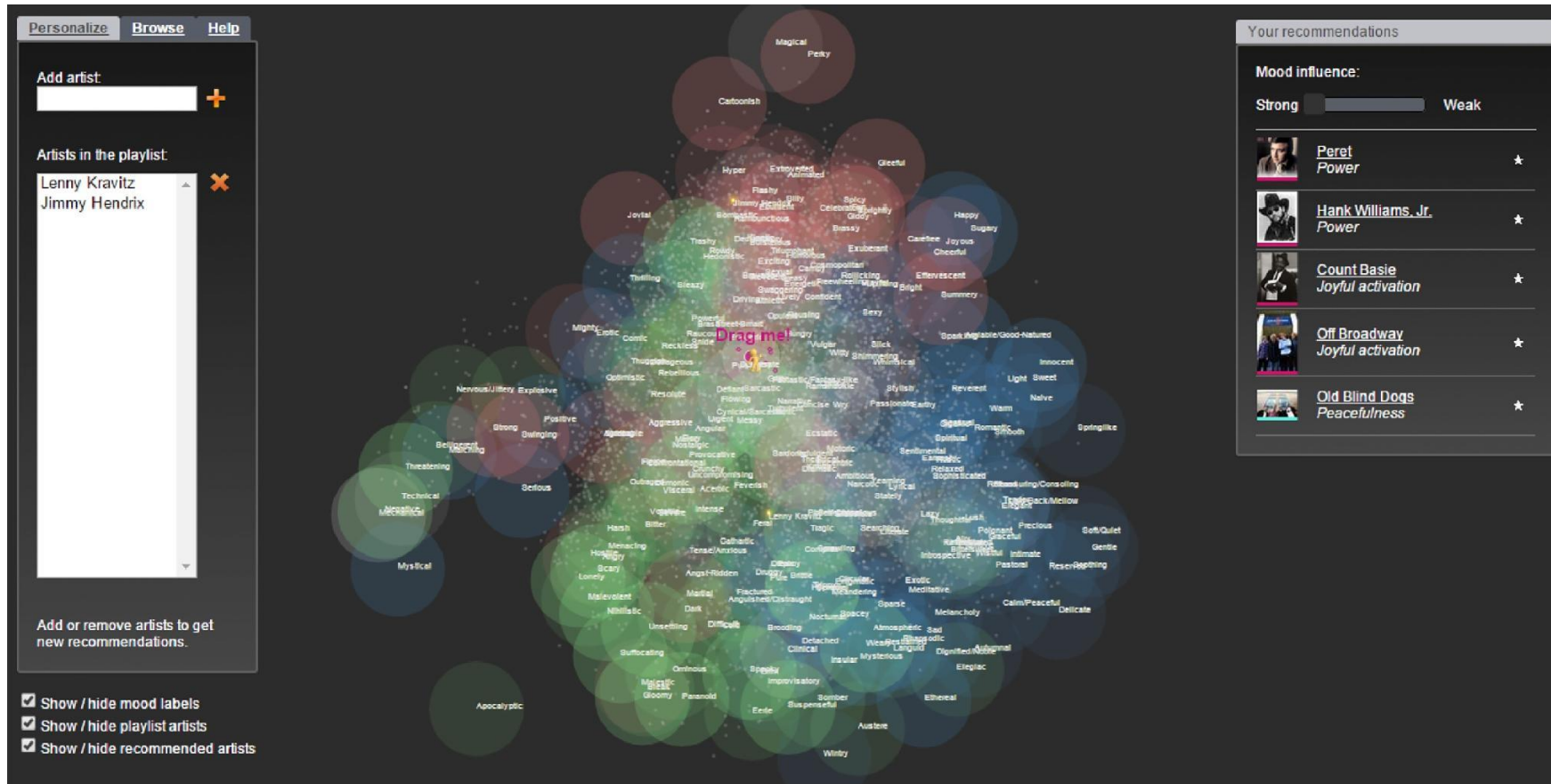
Adaptive VIBE: Relevance-Based Visualization



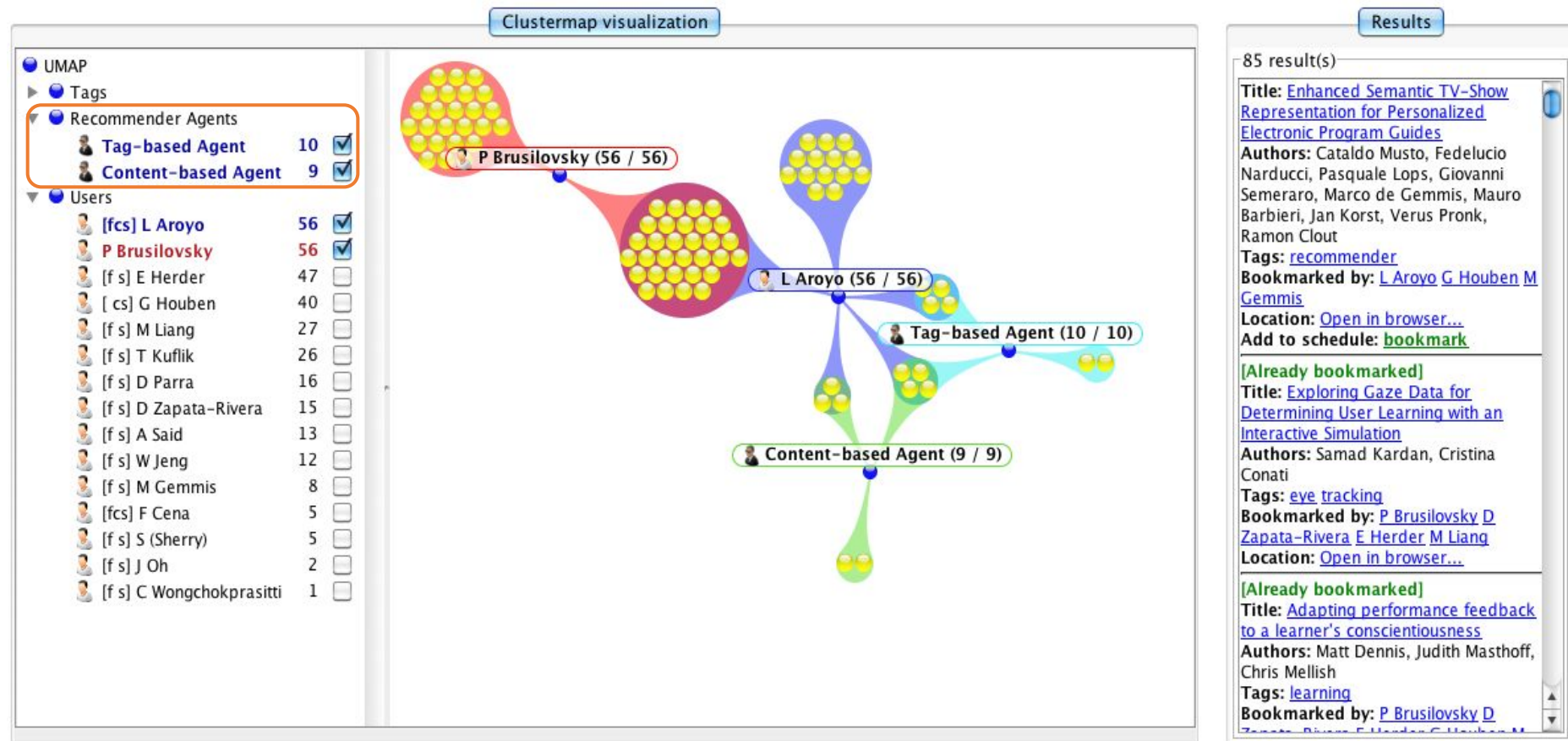
Adaptive VIBE: Relevance-Based Visualization



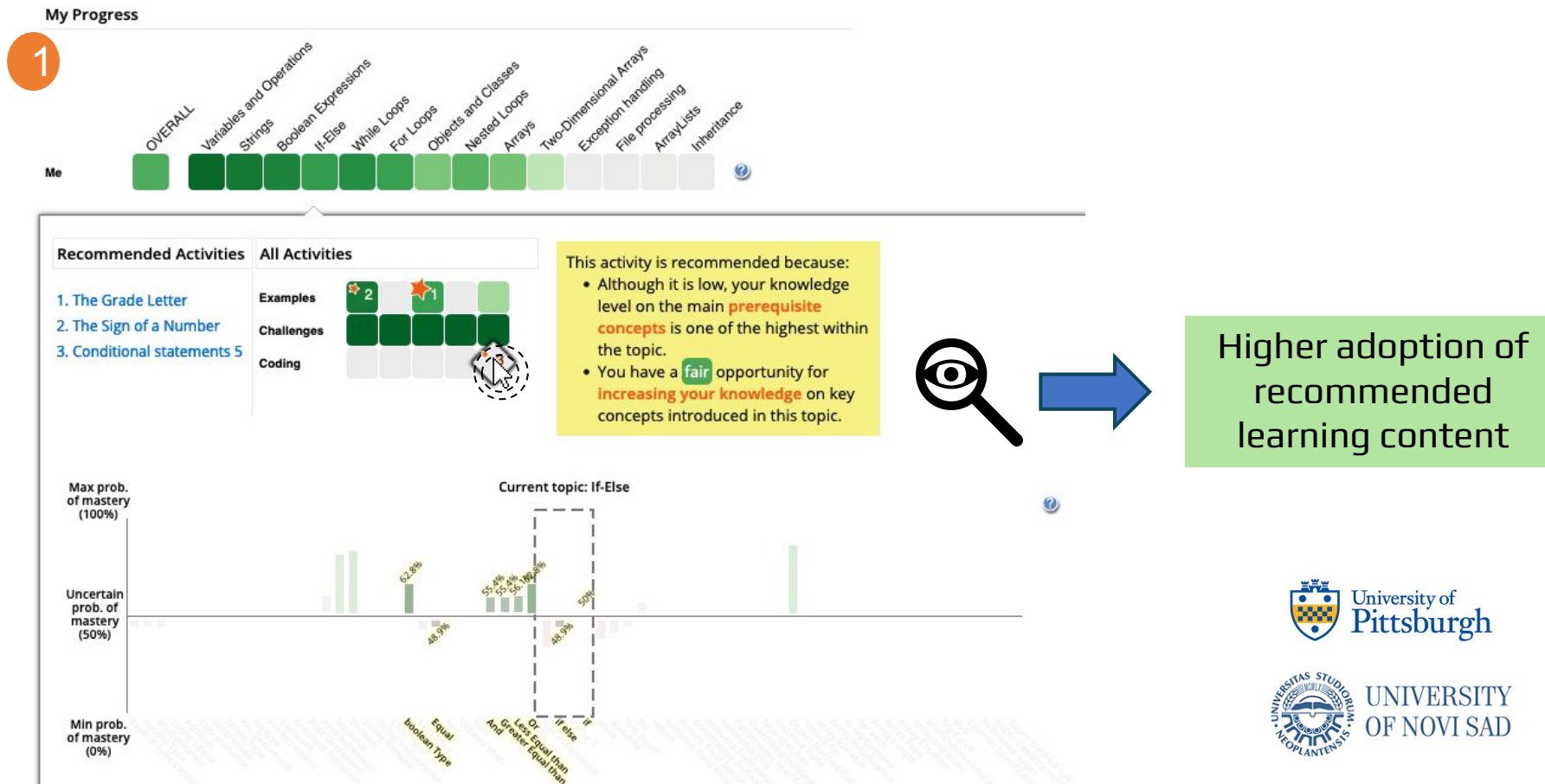
MoodPlay: Similarity-Based Visualization



TalkExplorer: Items to Users Connections (CF)

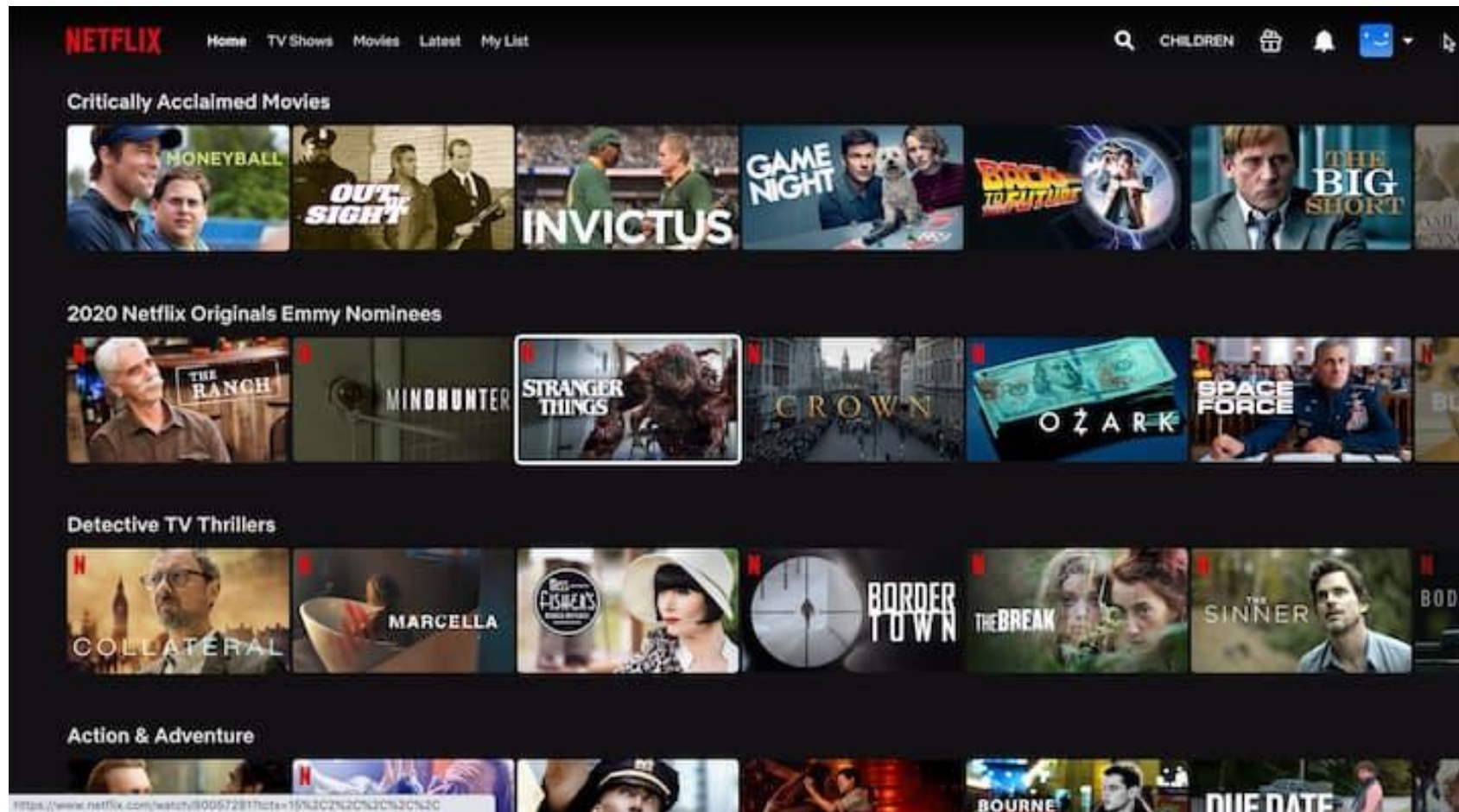


MasteryGrids: Item to Concept connections (CBF)



Barria-Pineda, J., Akhuseyinoglu, K., Želem-Čelap, S., Brusilovsky, P., Klasnja Milicevic, A., and Ivanovic, M. (2021) Explainable Recommendations in a Personalized Programming Practice System. In: 22nd International Conference on Artificial Intelligence in Education, AIED 2021, Springer, pp. 64-76.

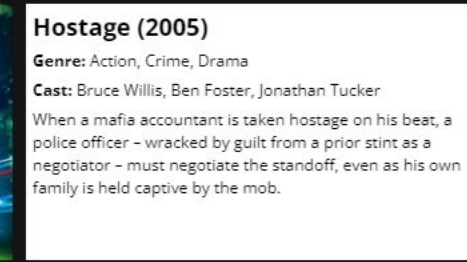
Carousel Presentation: Netflix “Navigator”





Interface-Aware Recommenders Through the Lens of Carousels

Crime



Animation



Romance

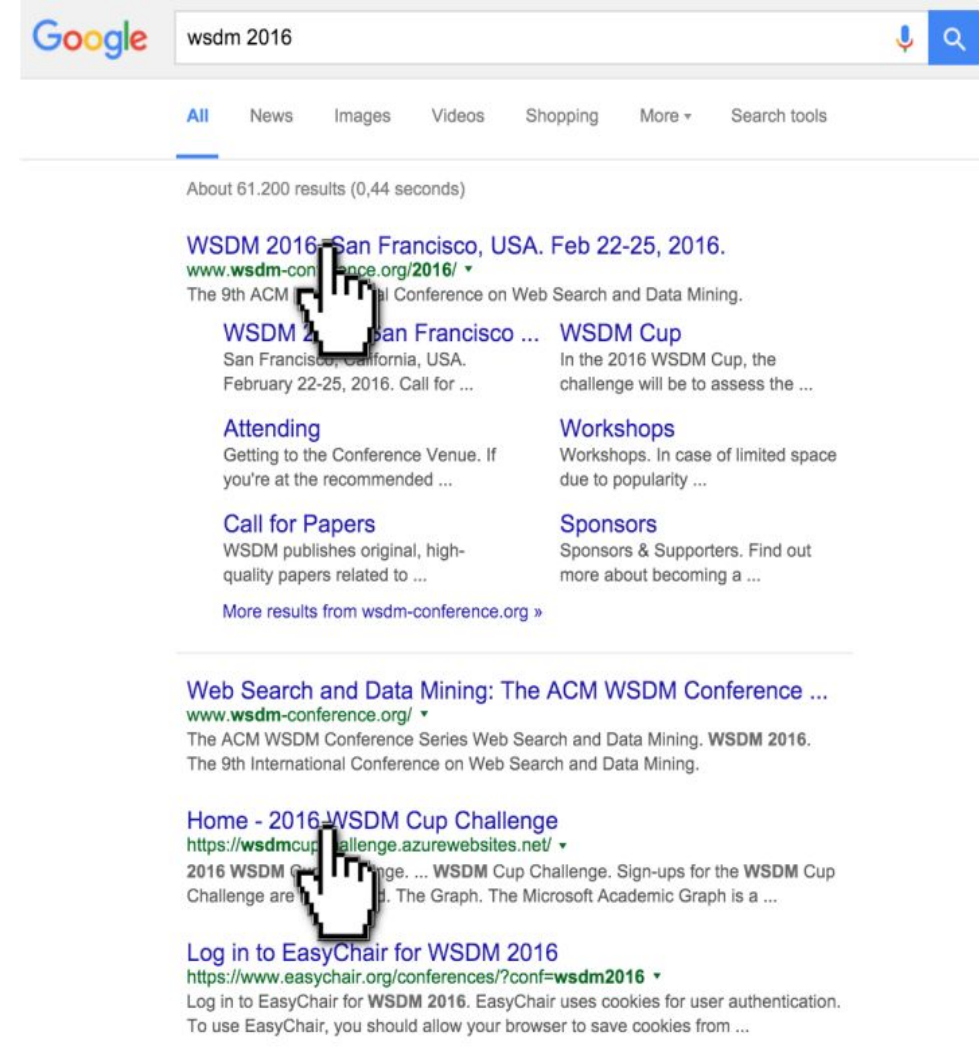




Challenges: Understanding How Users Browse and Interact

What is Browsing Behavior?

- Browsing Behavior
 - Everything that happens before the click
 - The sequence of actions/interactions taken by the user up until the final interaction (Click)

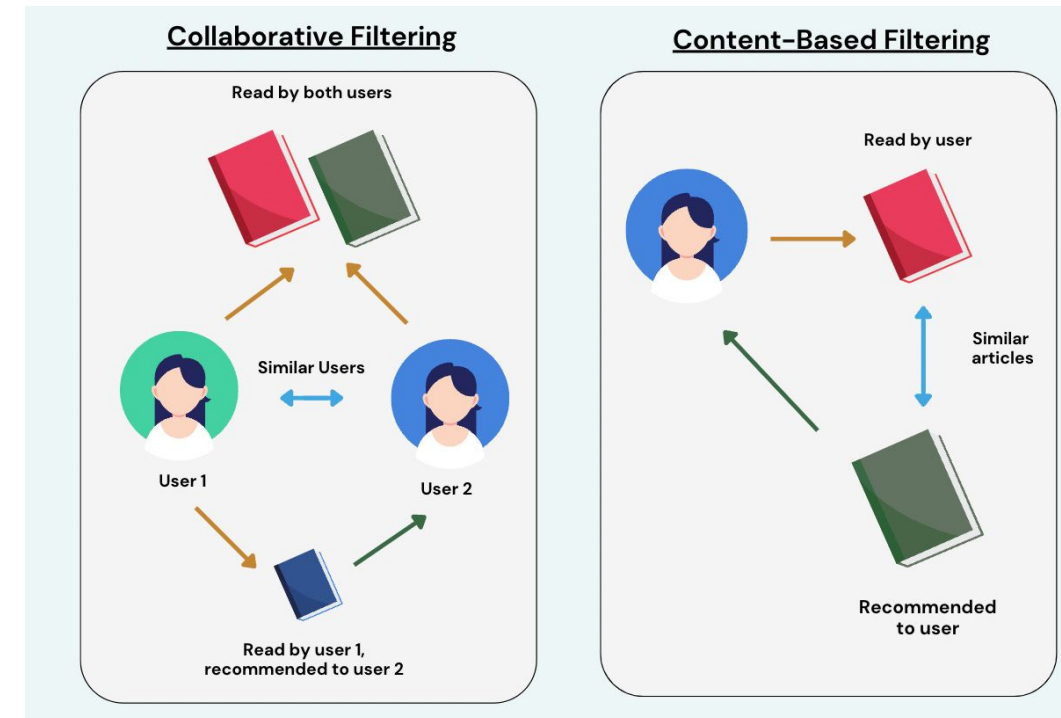


Why Browsing Behavior Matters?

- Impacts all the feedback you gather
- You are already taking it into account

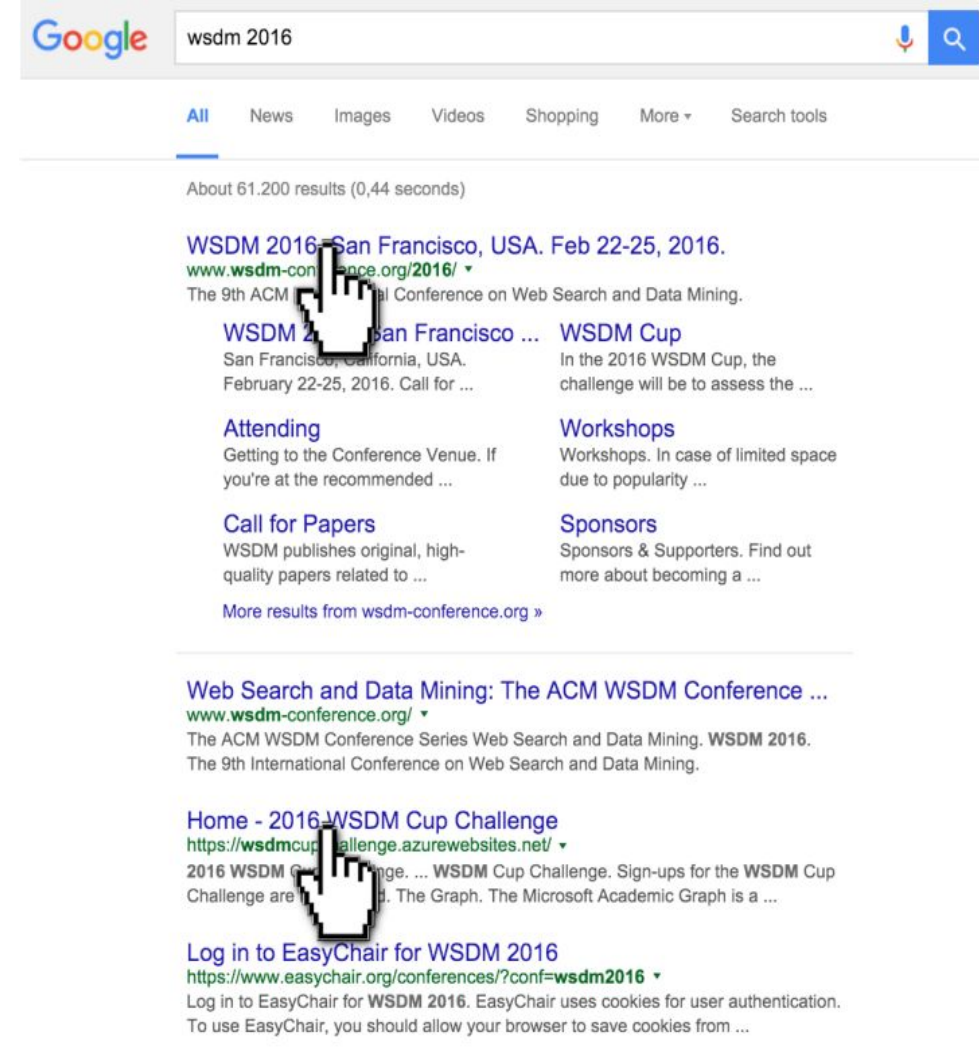
Why Browsing Behavior Matters?

- Browsing Behavior assumptions are in
 - Recommender Design Choices



Why Browsing Behavior Matters?

- Browsing Behavior assumptions are in
 - Recommender Design Choices
 - User/Click Models



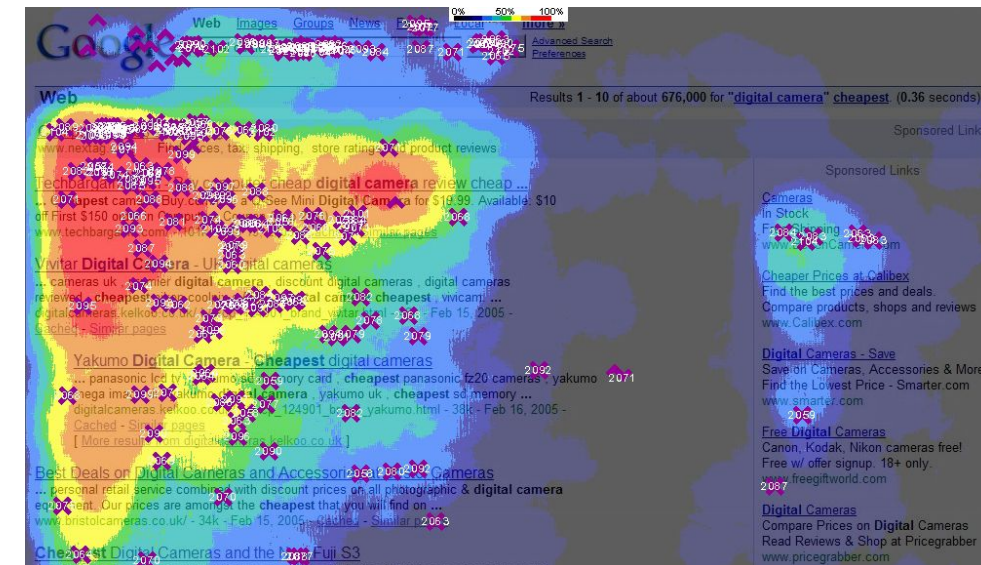
Why Browsing Behavior Matters?

- Browsing Behavior assumptions are in
 - Recommender Design Choices
 - User/Click Models
 - Evaluation Metrics

$$DCG(y) = \sum_{j=1}^J g(y_j) \cdot d(j) = \sum_{j=1}^J \frac{g(y_j)}{\log_2(j+1)},$$

Why Browsing Behavior Matters?

- Browsing Behavior assumptions are in
 - Recommender Design Choices
 - User/Click Models
 - Evaluation Metrics
 - UX or Item Presentation Decisions





This PC



older_tracker_script.txt



SJ-5011_TA-2_Export...



SJ-2001_TA-1_Export...



dexie console
commands.txt



i2 Testplayer



redme.rtf



Recycle Bin



current_script.txt



i2RecordingDesigner.e
xe - Verknüpfung



Preferences



SJ-2002_TA-1_Export...



reg_edits.txt



subj



New folder



export

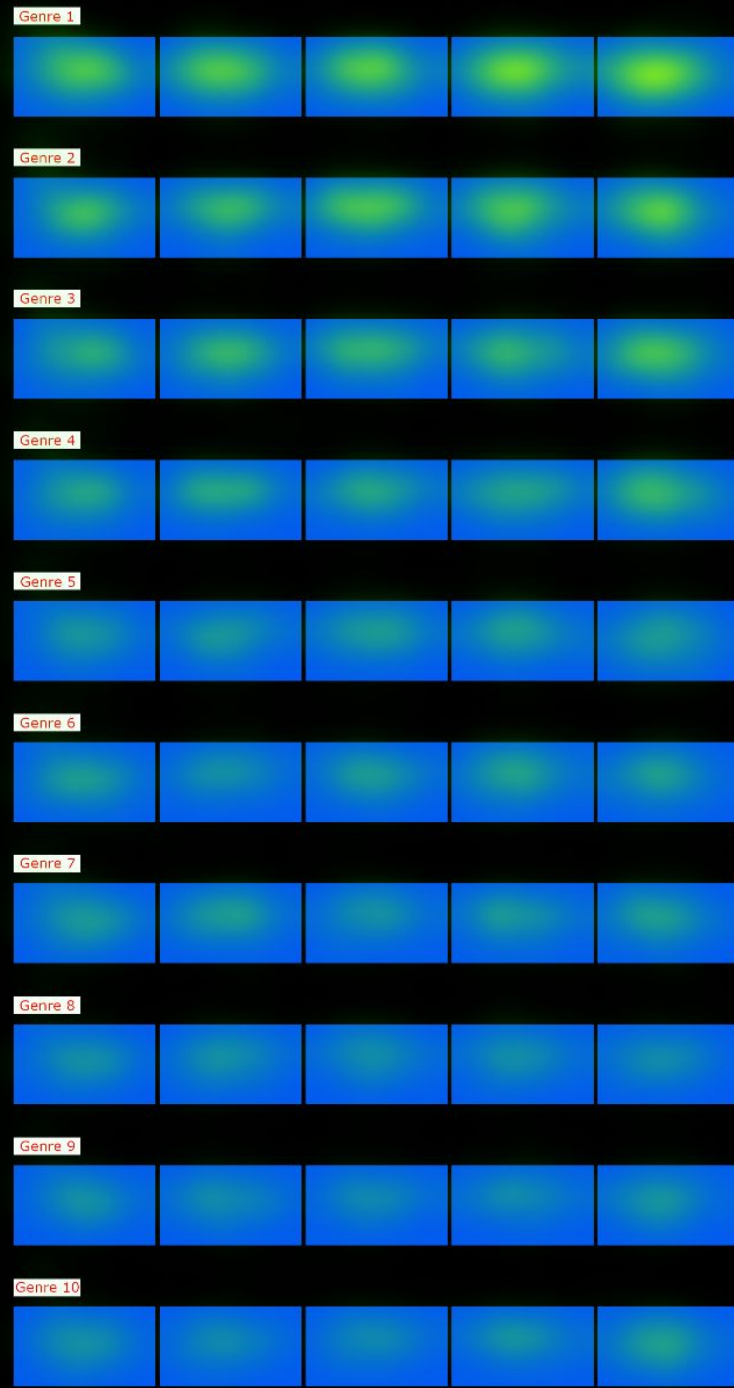
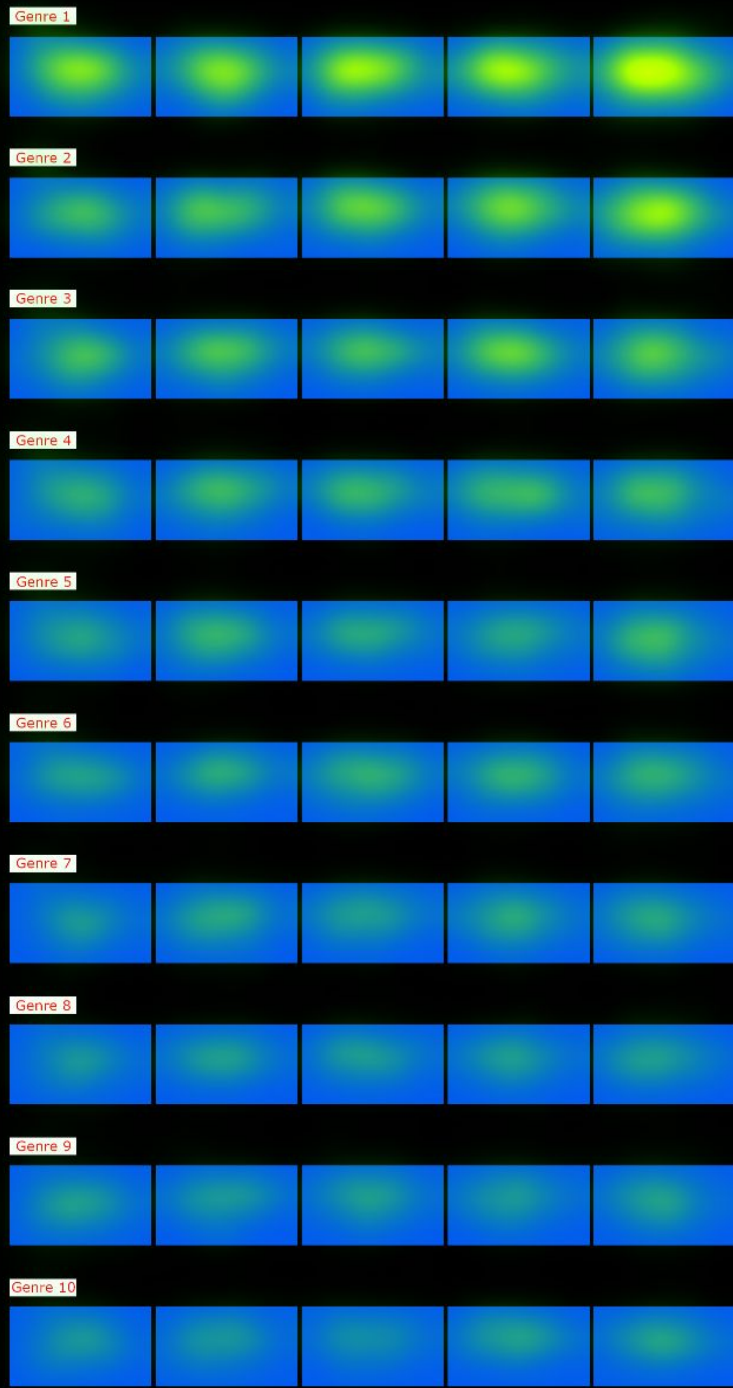
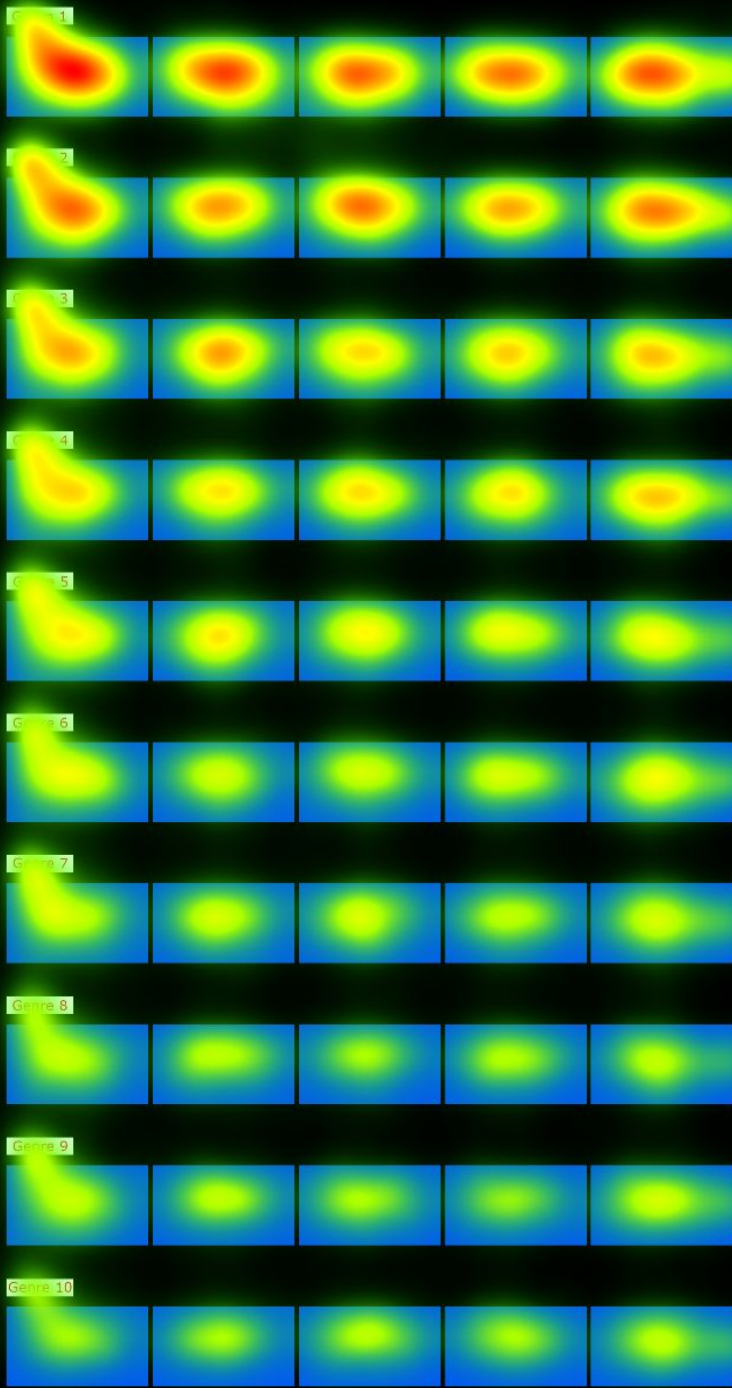


Search



ENG

1:30 PM



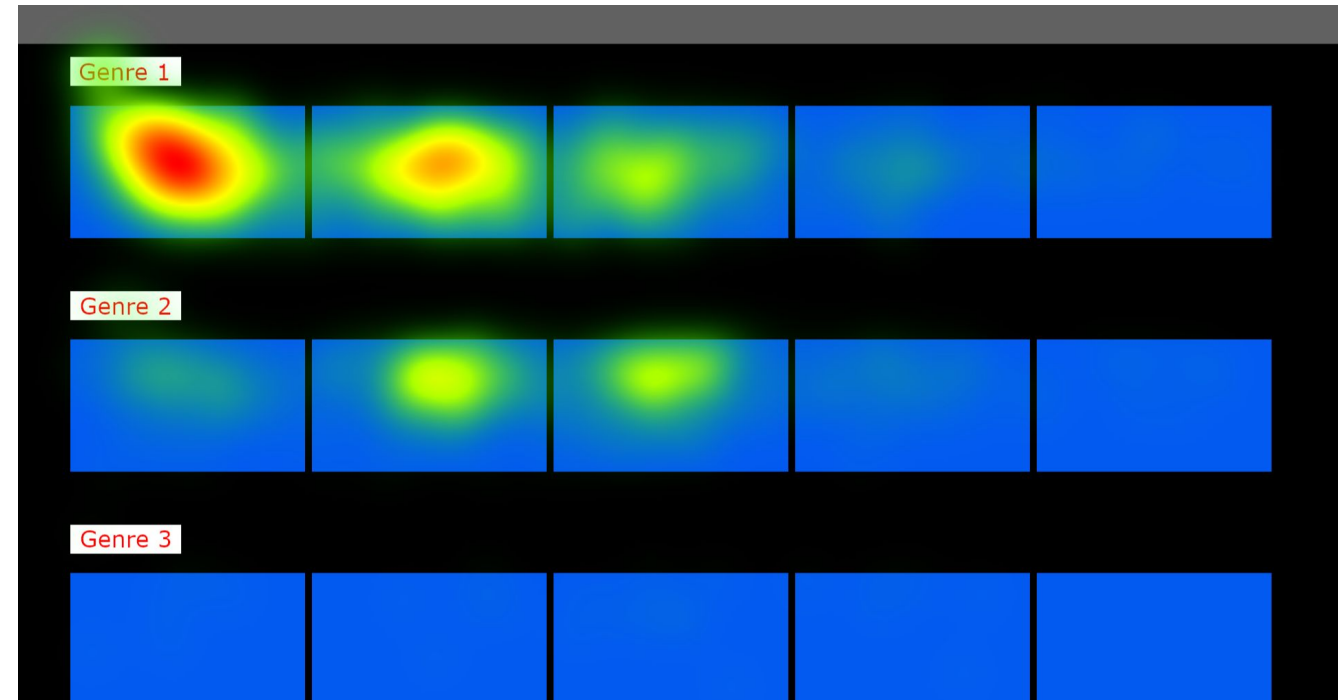
Why Browsing Behavior Matters?

- Browsing Behavior assumptions are in
 - Recommender Design Choices
 - User/Click Models
 - Evaluation Metrics
 - UX or Item Presentation Decisions

Standard Browsing (Initial/Unswiped Items)					Standard Browsing (Swiped Items)				
Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10
Initial Browsing Behavior (Initial/Unswiped Items)					Initial Browsing Behavior (Swiped Items)				
Item 1	Item 2	Item 3	Item 4	Item 5	Item 10	Item 9	Item 8	Item 7	Item 6

What can impact browsing behavior (and clicks)?

- Position Bias



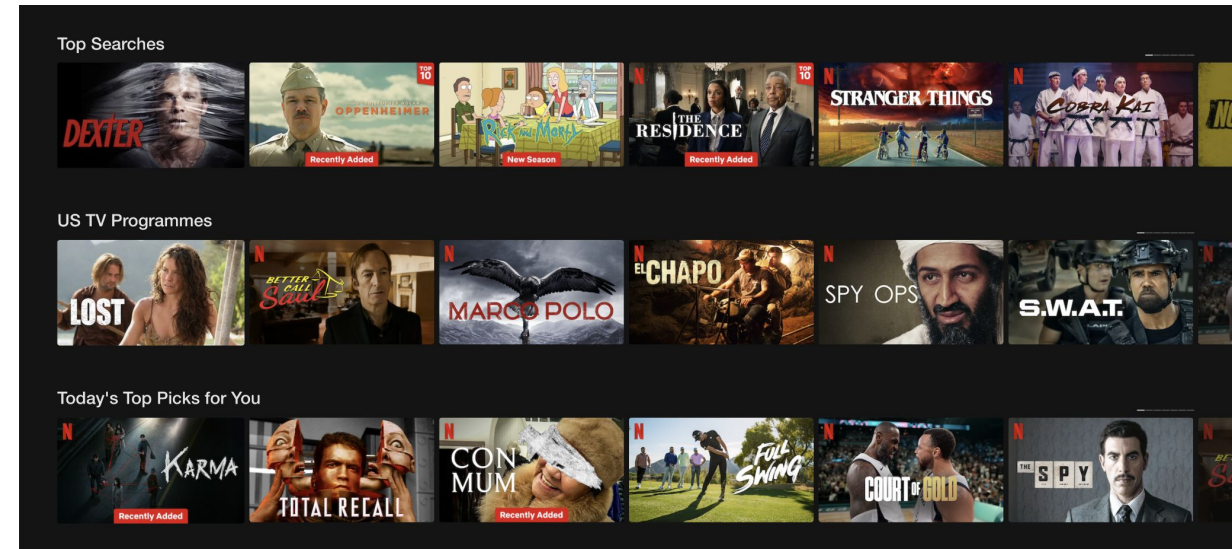
What can impact browsing behavior (and clicks)?

- Position Bias
- Visual Bias (Posters, text, etc.)



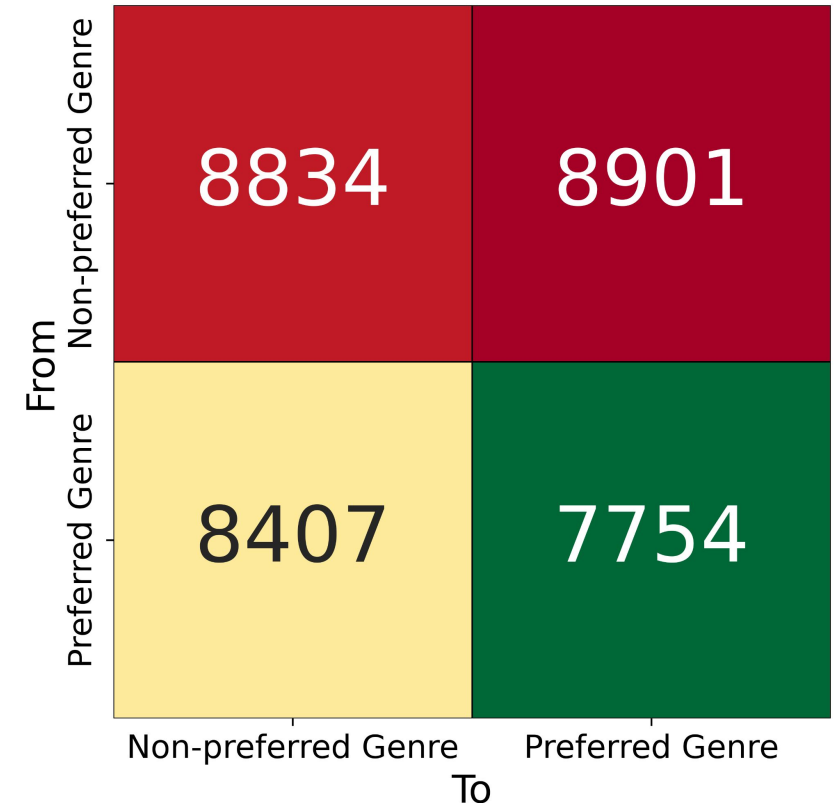
What can impact browsing behavior (and clicks)?

- Position Bias
- Visual Bias (Posters, text, etc.)
- Topics/Genres



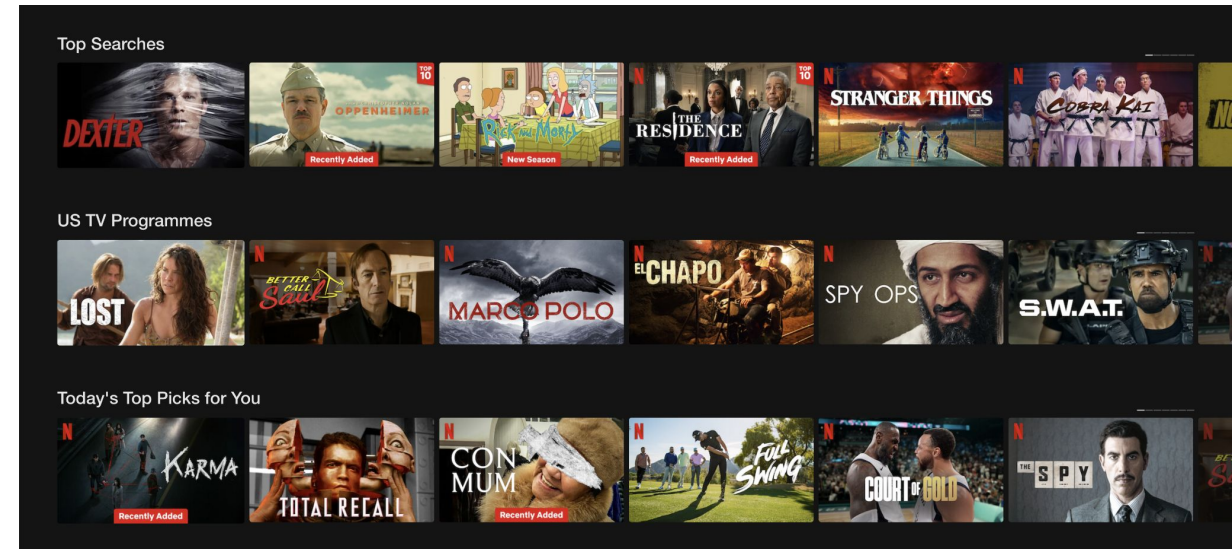
What can impact browsing behavior (and clicks)?

- Position Bias
- Visual Bias (Posters, text, etc.)
- Topics/Genres
 - Preferred to Non-preferred: 7,754
 - Non-preferred to Preferred: 8,901
 - Wilcoxon signed-rank test ($w=3205.5$, $p<0.001$)

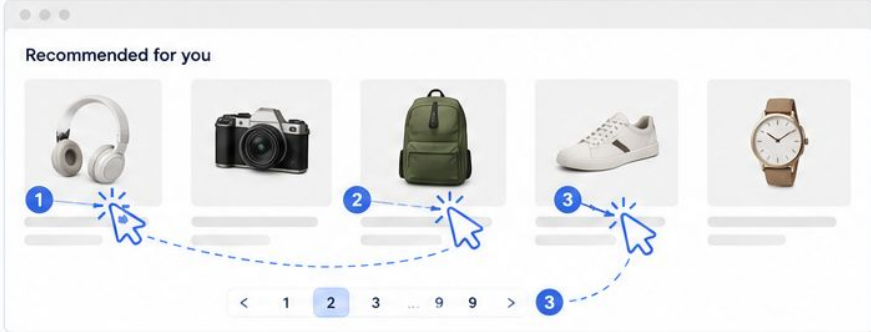


What can impact browsing behavior (and clicks)?

- Position Bias
- Visual Bias (Posters, text, etc.)
- Topic/Preference Bias
- Others:
 - User Context (Needs, Query)
 - Trust Bias
 - Item Selection Bias
 - Surrounding Item Bias



How to measure browsing behavior?

LEVEL	EXAMPLE VISUALIZATION	EASE & INFORMATIVENESS
LEVEL 1 Clicks 		 Easy to gather, but not very informative. Shows what users clicked, but not what captured attention or why.
LEVEL 2 Cursor Movement 		 Moderate to gather, more informative. Reveals interest and exploration patterns, even without clicks.
LEVEL 3 Eye Tracking 		 Harder to gather, most informative. Shows where and how users visually focus, revealing attention, processing, and cognitive effort.

Summary

- Browsing behavior is an integral part of the recommender loop
- Eye tracking is the gold standard, but costly



Future Directions in Browsing Behavior

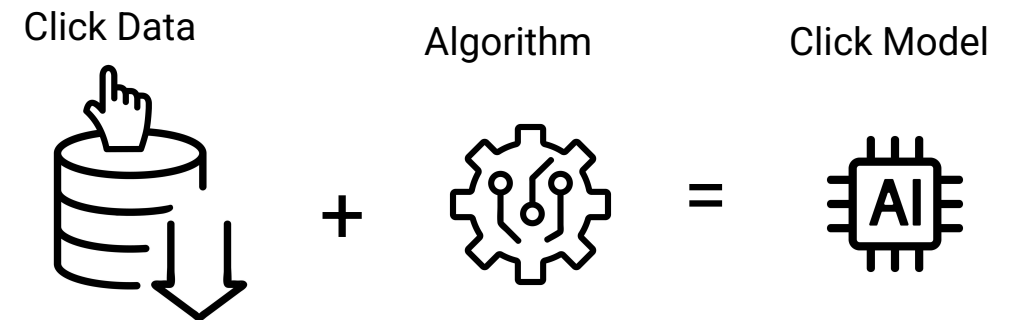
- Eye tracking studies in different settings
- Sequential analysis of eye tracking or cursor data
- Browsing behavior differences between users
- Cursor a good enough proxy for eye tracking?



Challenges: Building Interface-Aware User Models

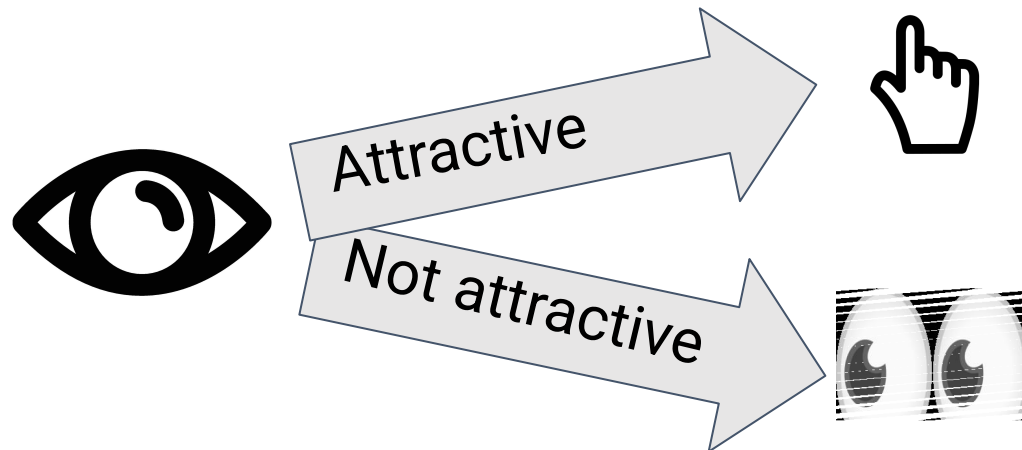
What is a click model?

- Probabilistic (computational) model that learns how people click
- How can we use click models?
 - Understanding users
 - Simulating users
 - Learning item relevance/attraction
 - Evaluating recommendations



General Click Model

- For an item to be clicked:
 - The position must be examined
 - Once it is examined the user must find the item attractive
- $P(\text{Click}) = P(\text{Examining the position}) * P(\text{Item is Attractive})$

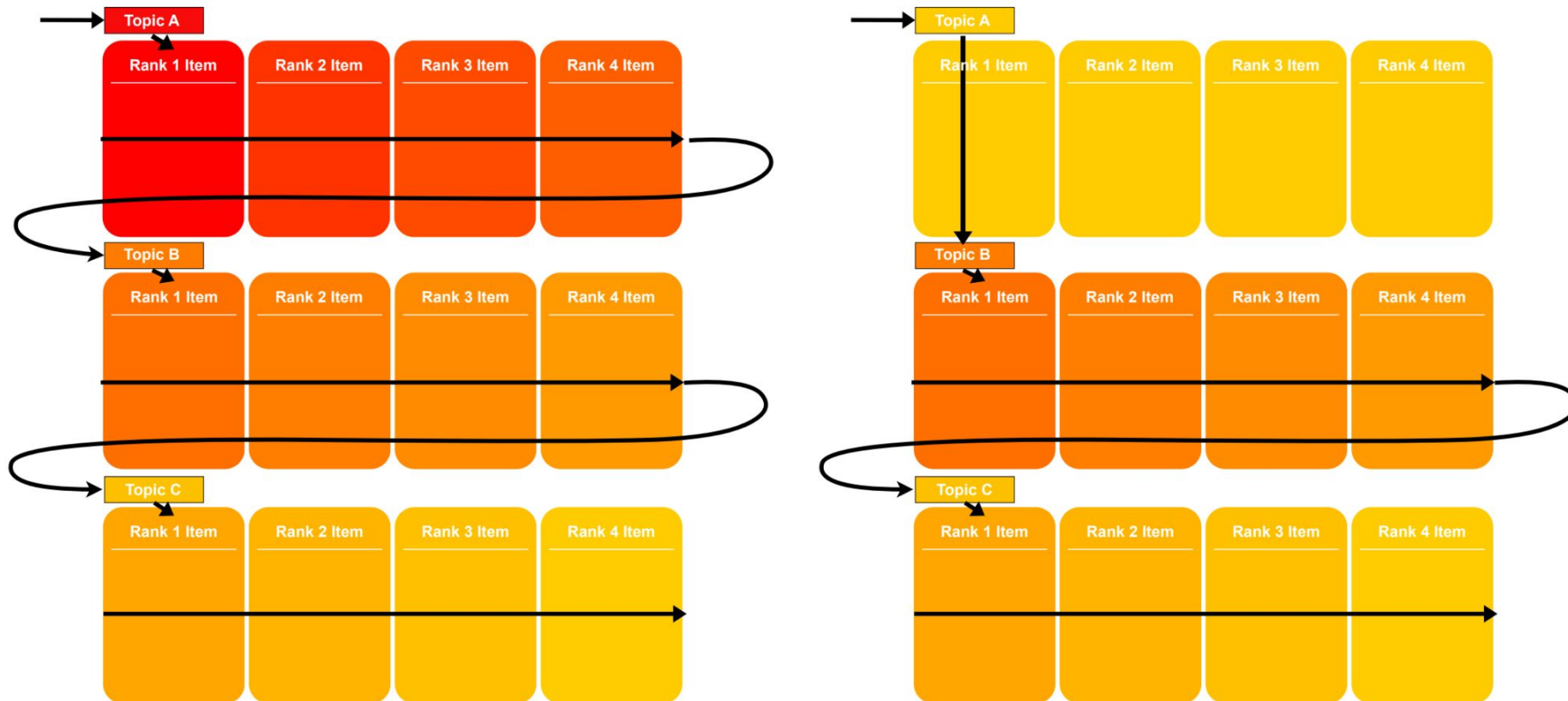


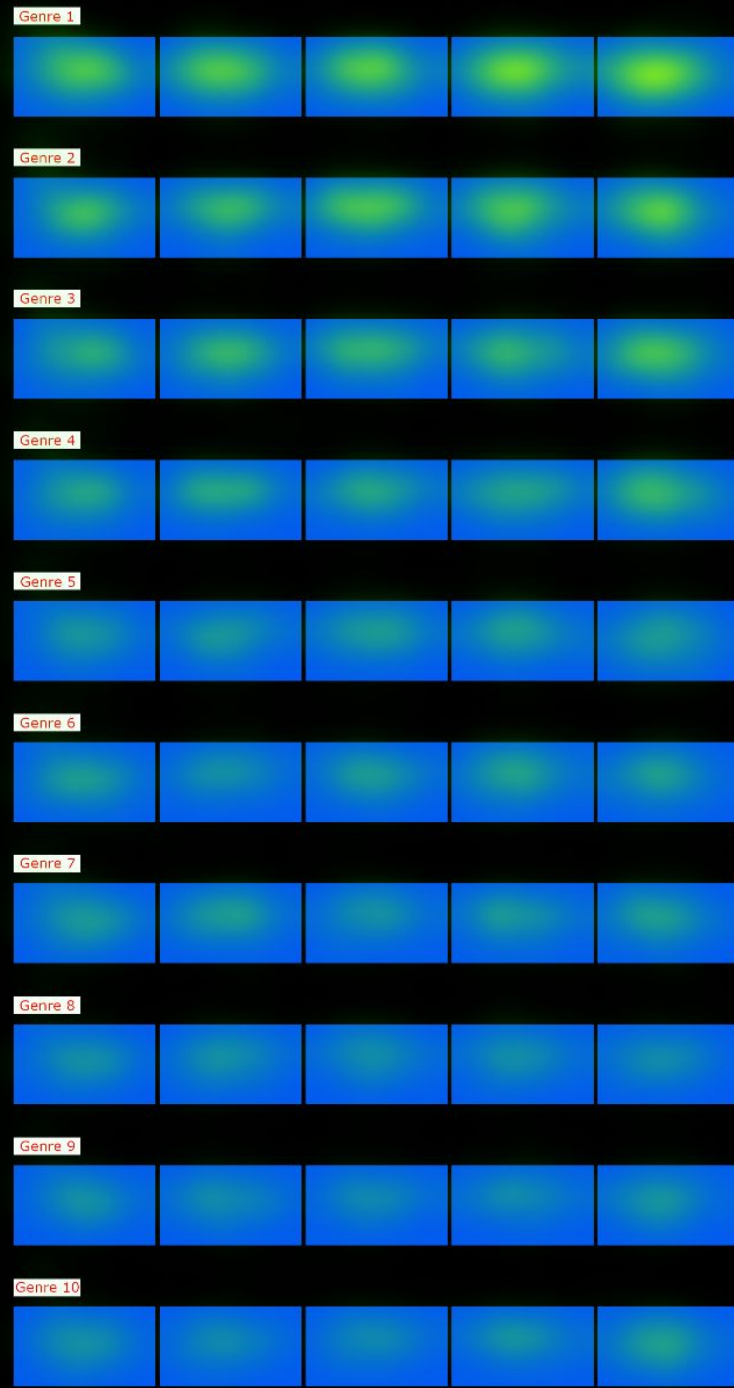
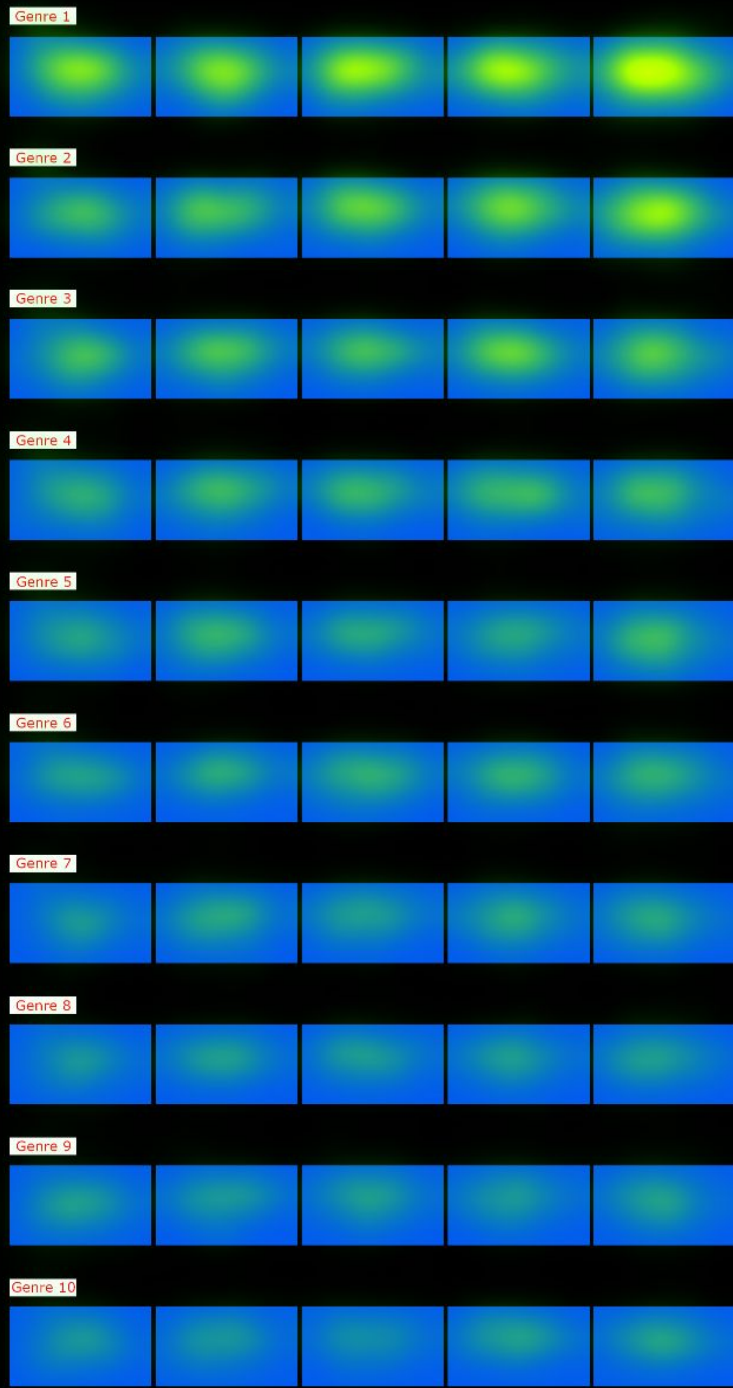
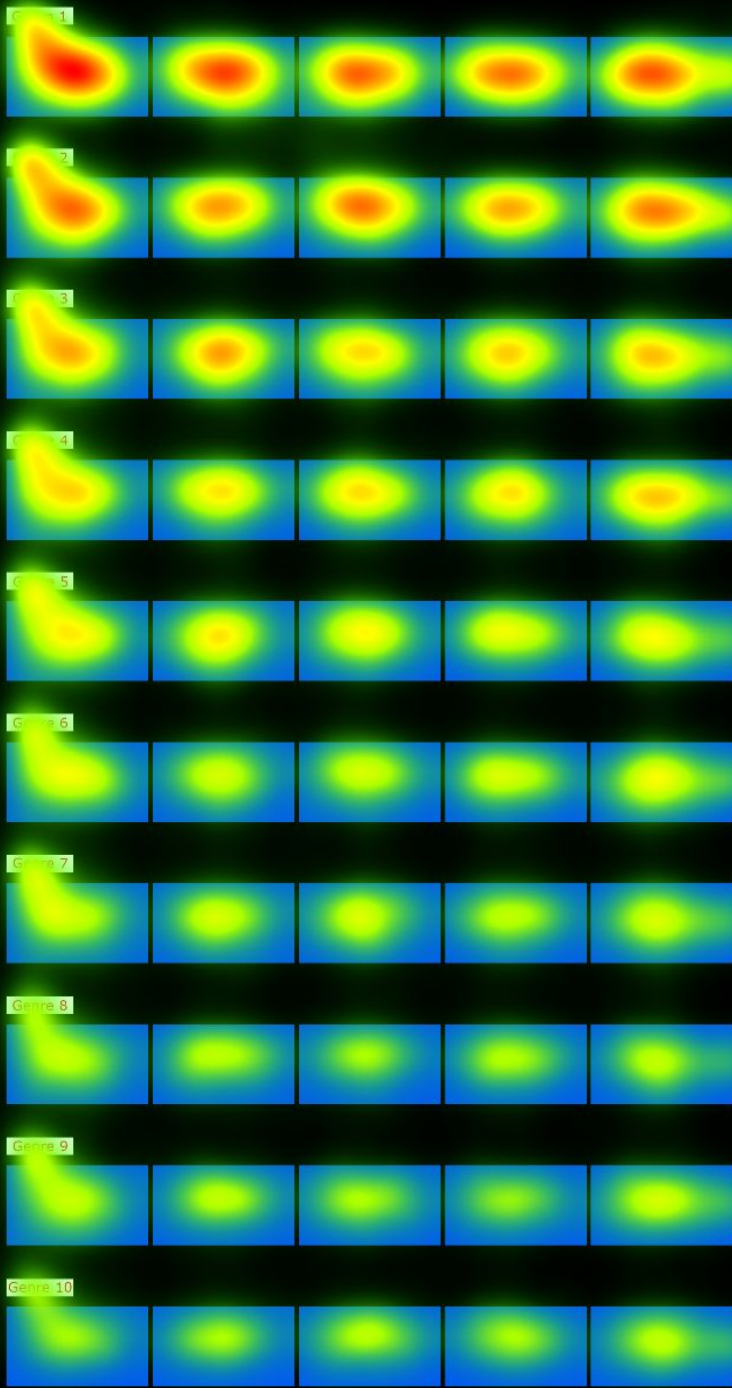
Types of Click models

- Position-based Model (PBM)
 - $P(\text{Click}) = P(\text{Examining the position}) * P(\text{Item is Attractive})$
- Other click models:
 - Change examination probability
 - Make assumptions about when things are examined

First Carousel Click Model:

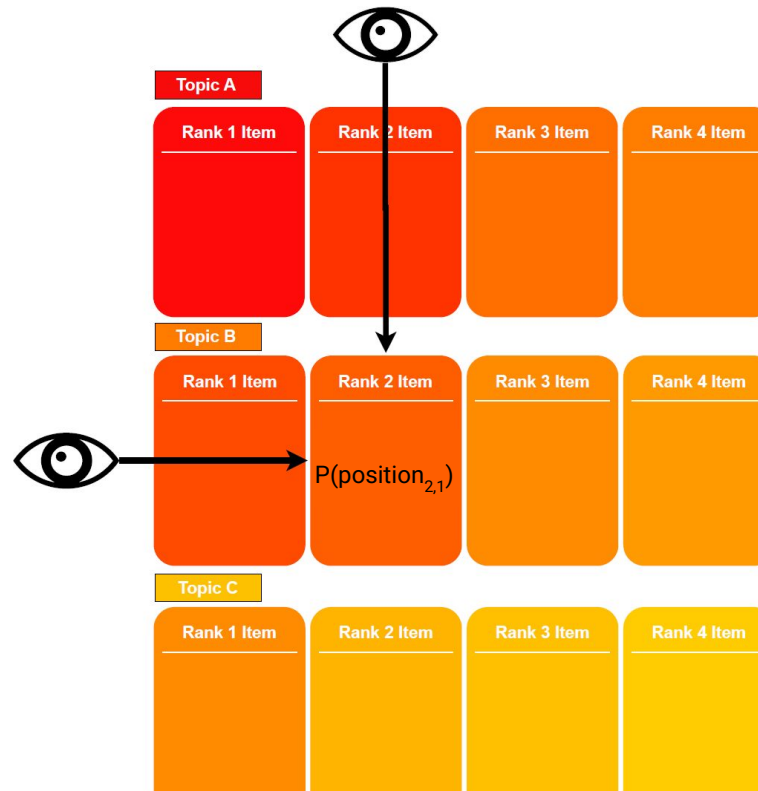
Carousel Click Model (CCM)





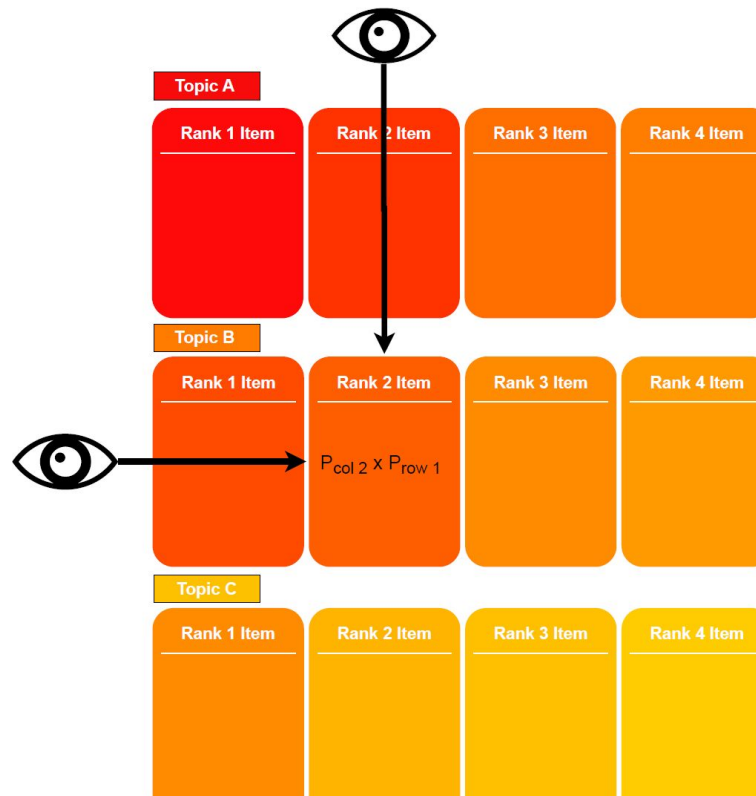
Carousel Position-Based Model (CPBM)

$$P(\text{click}) = P(\text{Examining the position}) * P(\text{Item is attractive})$$



Row Column Position-Based Model (RCPBM)

$$P(\text{click}) = P(\text{Examining Row}) * P(\text{Examining Column}) * P(\text{item is attractive})$$



Problem in click models: Learning two things

$$P(\text{Click}) = P(\text{Examining the position}) * P(\text{Item is Attractive})$$

- Learn attraction
 - $P(\text{Item is attractive})$
 - Learned from an observable signals (clicks)
- Learn examination
 - $P(\text{examining a position})$
 - **Unobserved (latent variable)**

Using Gaze for the Observed Examination Position-Based Model (OEPBM)

Standard Click Log-Likelihood

$$\mathcal{LL} = \sum_{t \in D} \boxed{c_t} \cdot \log(w_t \theta_t) + \boxed{(1 - c_t)} \cdot \log(1 - w_t \theta_t)$$

Clicked

Not Clicked

Observed Examination Log-Likelihood (OELL)

$$\mathcal{LL} = \sum_{t \in D} \boxed{c_t e_t} \cdot \log(w_t \theta_t) + \boxed{(1 - c_t) e_t} \cdot \log(w_t (1 - \theta_t)) + \boxed{(1 - c_t)(1 - e_t)} \cdot \log(1 - w_t)$$

Clicked &
Examined

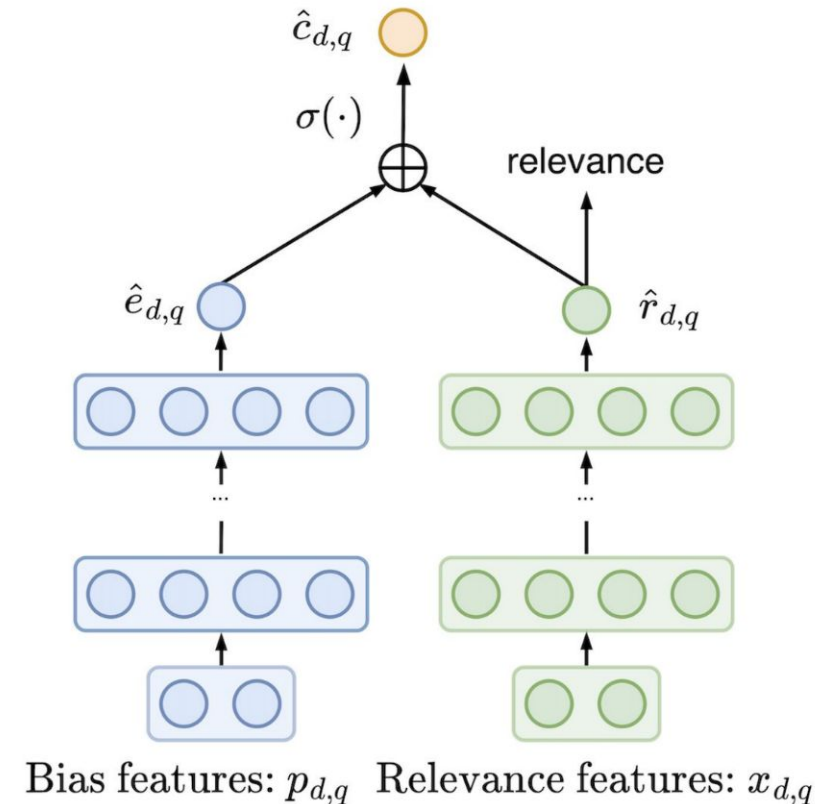
Not Clicked &
Examined

Not Clicked &
Not Examined

Click Models as Rankers: Two Tower Models

Two Tower models are Neural Contextual PBMs

- Relevance tower: query-document features
- Bias Tower: contextual features
 - position, dwell time, impressions



Summary

- Interface-aware systems leverage browsing behavior insights
- Requires interface-specific user models

Future Directions in User Modeling

- Gaze-based user models or recommender systems
- Contextual user models (e.g, take into account preference)
- Exploration of other feedback (e.g., dwell time, cursor, visits)
- New click models for multi-lists (e.g. Neural click models)



Challenges: Interface-Aware Evaluation

Evaluating in a Multi-list setting

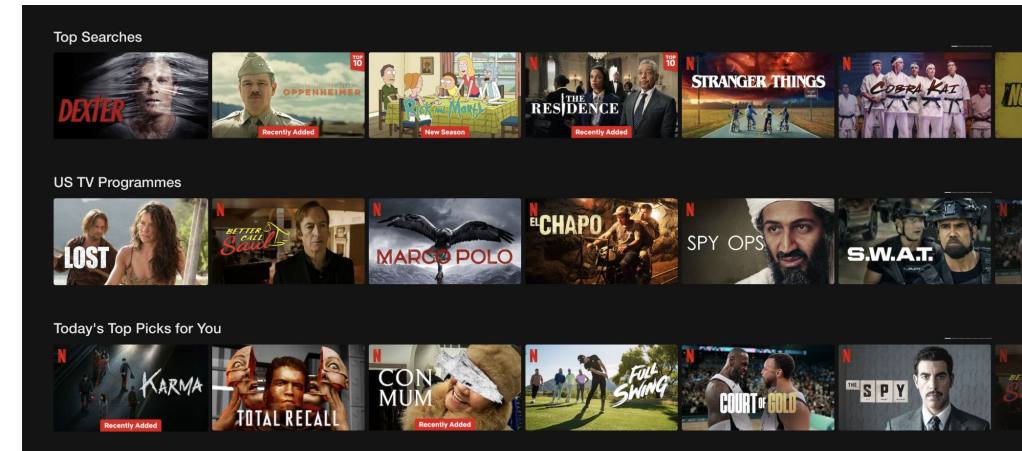
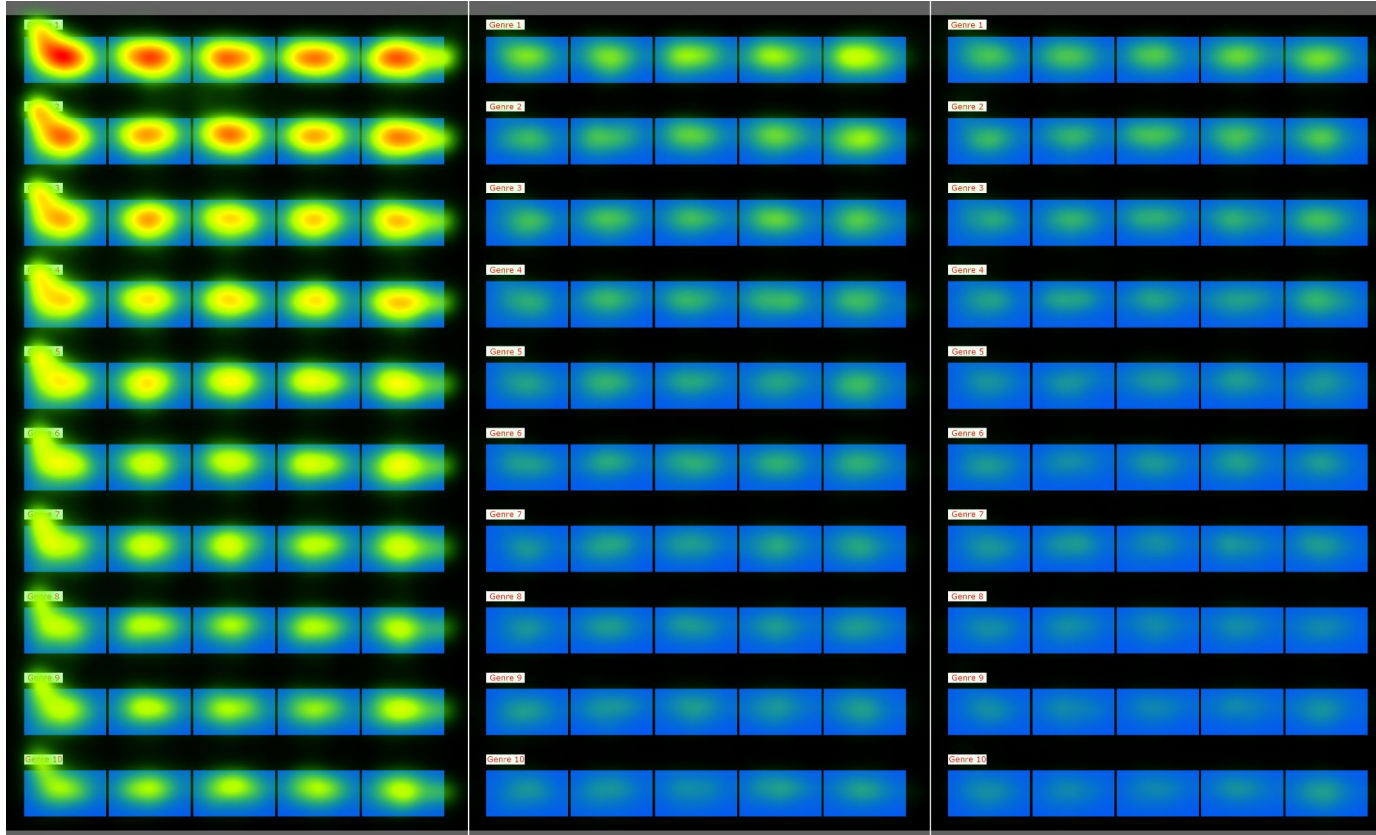
Problem: How do we evaluate over multiple lists?

- Single-list optimization
- Whole-page (Multi-list) optimization

Problems with list-based ranking metrics

- Non-rank order dependent - consider whole list
 - precision, recall, AUC
- Rank order dependent - consider part of the list or care about ordering
 - @N metrics, Mean Reciprocal Rank, NDCG,

Problem: Finding the Top-N



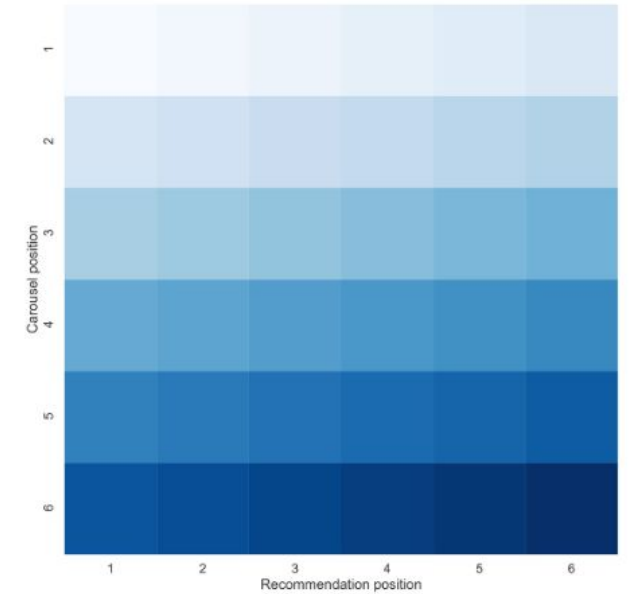
Problem: DCG order dependence

$$2DCG(y) = \sum_{i=1}^I \sum_{j=1}^J g(y_{i,j}) \cdot d(i, j),$$

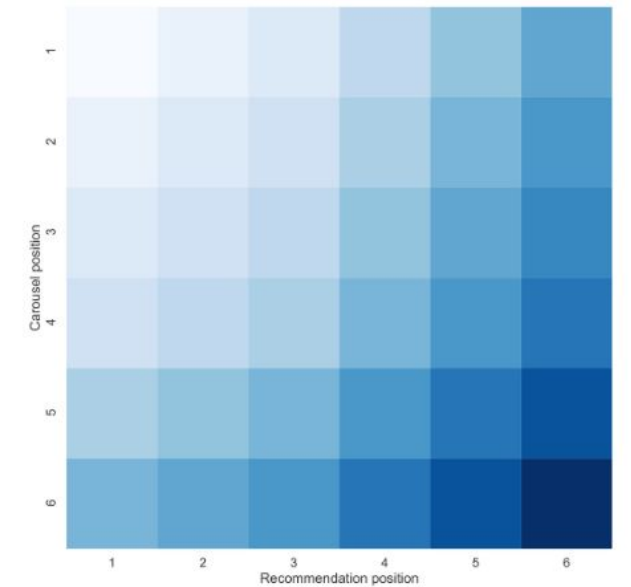
Ranking
Gain or Relevance Function
Position Discount Function

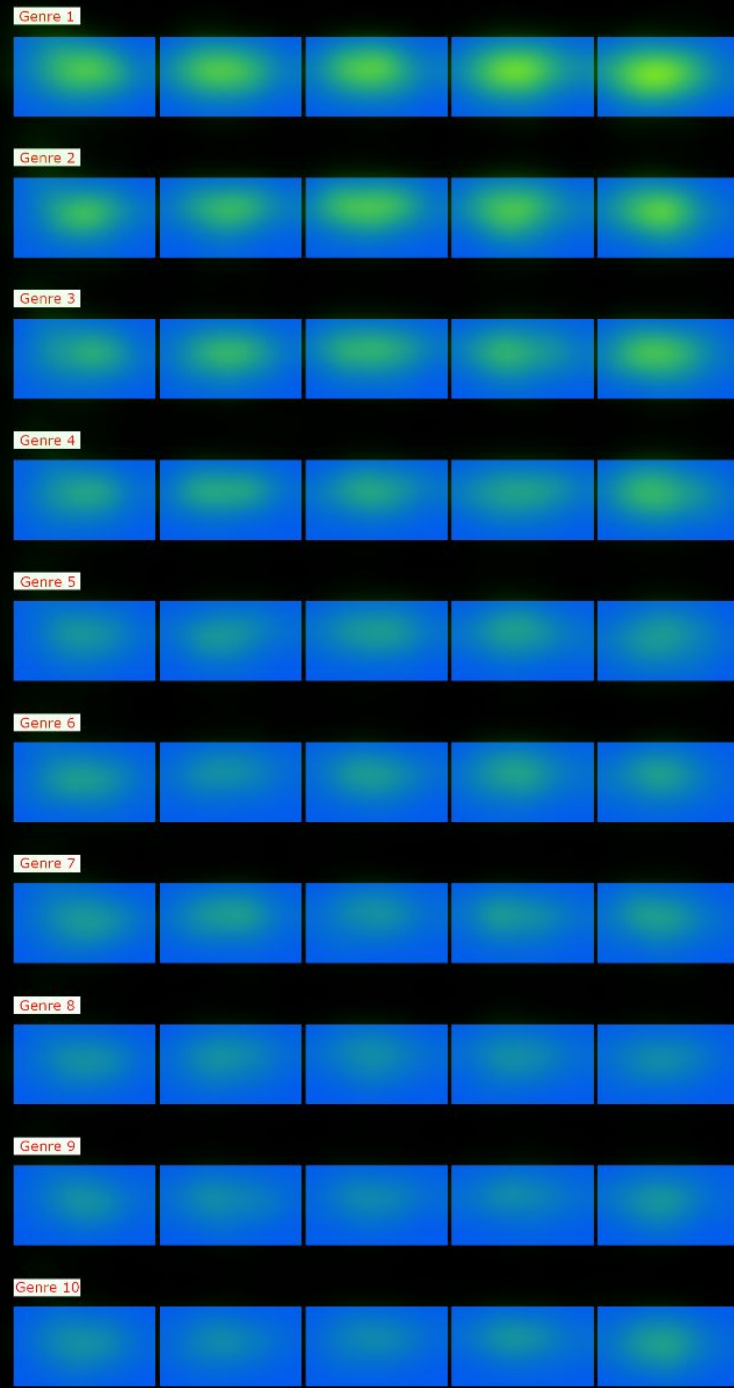
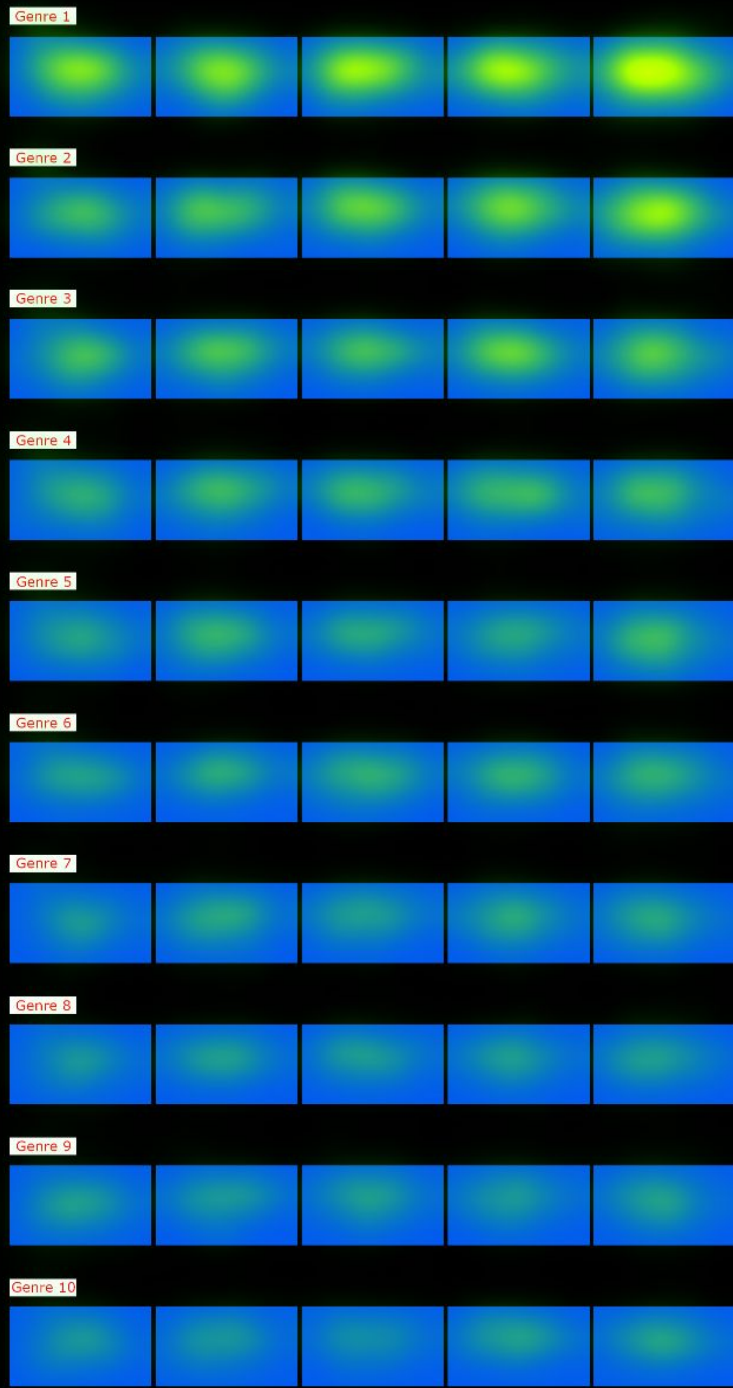
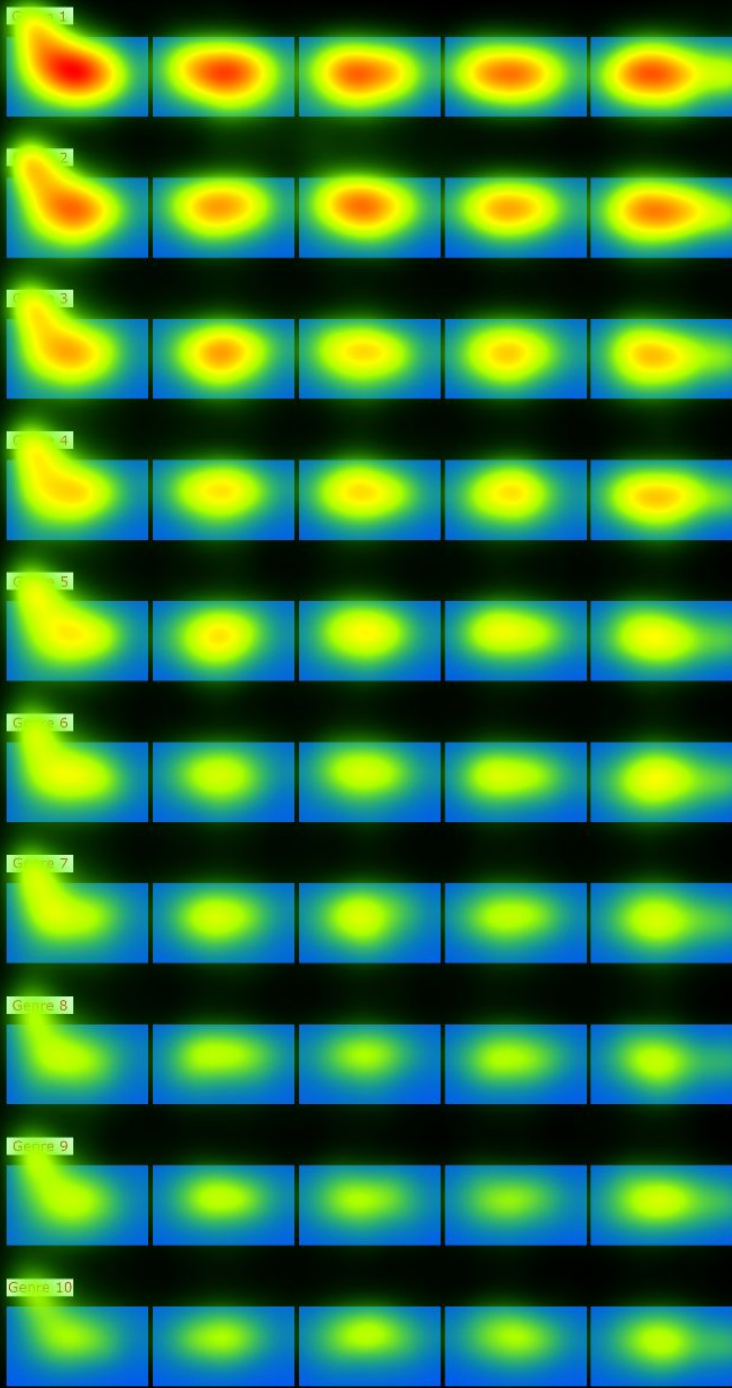
$$d_t(i, j) = \frac{1}{\log_2(\alpha i + \beta j)}, \quad d_a(i, j) = \frac{1}{\log_2(\alpha i + \beta j + \gamma n_h + \lambda n_v)}$$

standard DCG



F-pattern 2DCG





Summary

- Evaluation metrics have browsing behavior assumptions
- Requires interface specific evaluation metrics

Future Directions in Evaluation

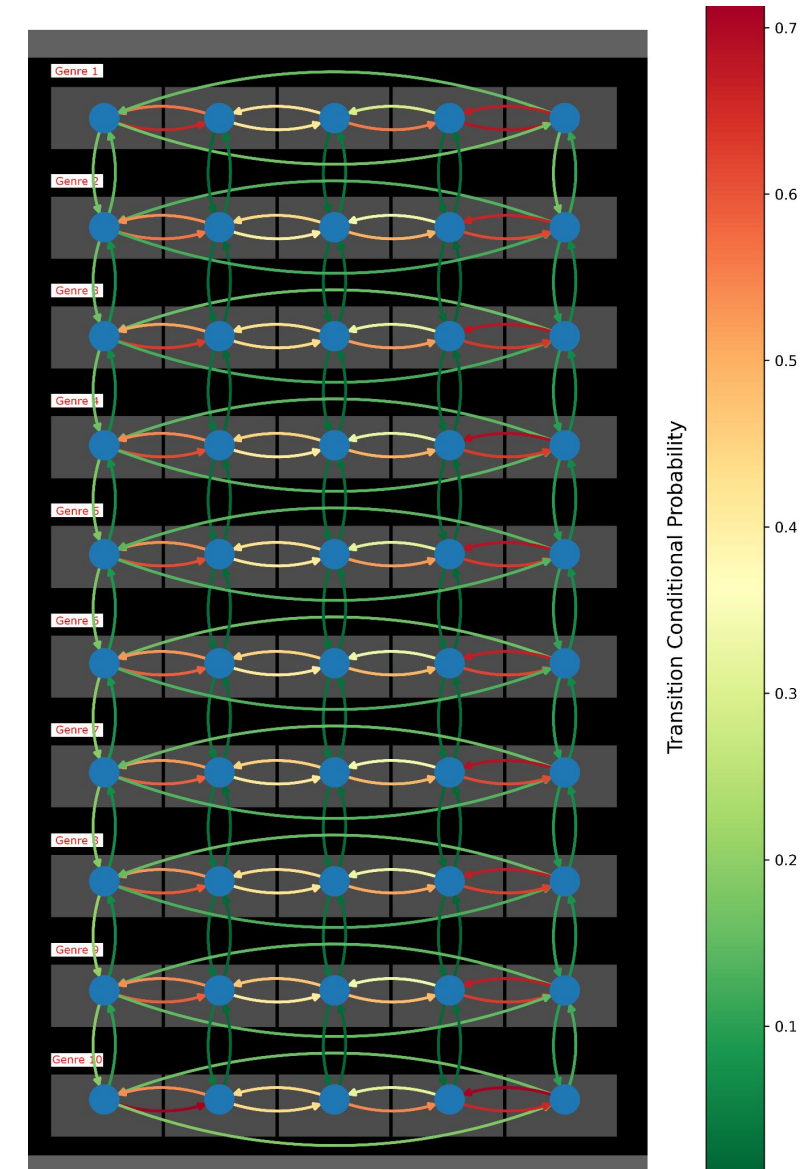
- How to deal with duplicate items?
- Inter-list interactions for evaluation
- Are adapted metrics inline with downstream metrics (e.g, recommendation quality)



**Discussion: What
interface specific
challenges you have
encountered?**

Discussion: Interface specific challenges or insights you have encountered?

- Carousels
 - Swiping Behavior
 - Top 2 row bias
 - Row skipping is rare
 - Topic Preference





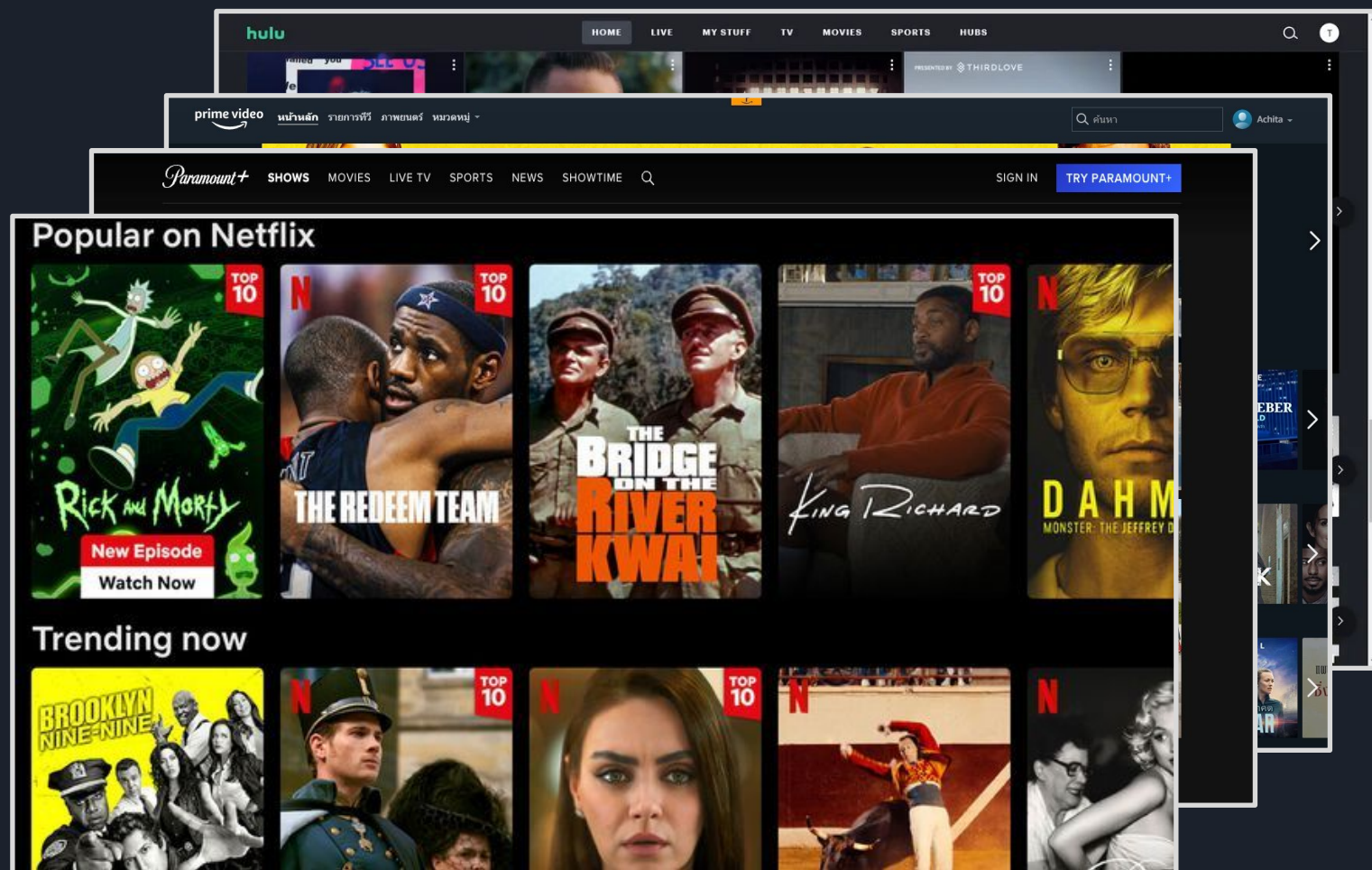
CARE: An Infrastructure for Evaluation of Carousel-Based Recommender Interfaces

Behnam Rahdari, Peter Brusilovsky
School of Computing and Information, University of Pittsburgh

CARE: An Infrastructure for Evaluation of Carousel-Based Recommender Interfaces

Carousel UIs shape discovery, navigation, and engagement.

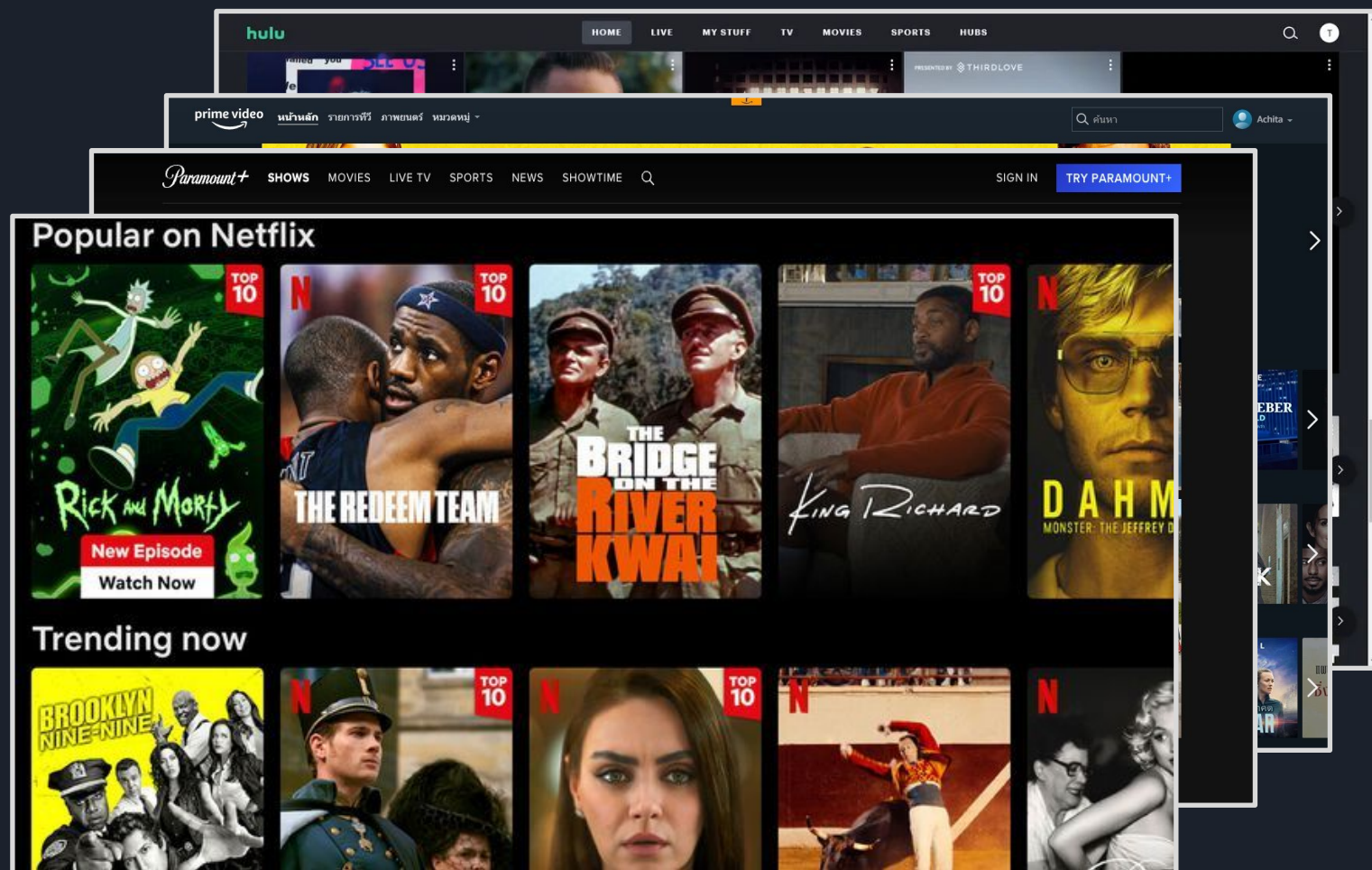
- Interface behavior is not captured by accuracy metrics alone.
- Carousel navigation creates unique interaction patterns.
- Researchers need comparable logs and reusable experiment setup.



CARE: An Infrastructure for Evaluation of Carousel-Based Recommender Interfaces

Carousel UIs shape discovery, navigation, and engagement.

- Interface behavior is not captured by accuracy metrics alone.
- Carousel navigation creates unique interaction patterns.
- Researchers need comparable logs and reusable experiment setup.



What is CARE?

CARE is an web-based platform that allow researchers to monitor studies, manage participant, and download interaction logs from a central dashboard.

- Study management and launch flow
- Configurable interaction settings
- Granular event capture for behavioral analysis

CarE: Carousel Evaluation Framework

Experiment Center

Experiment Center

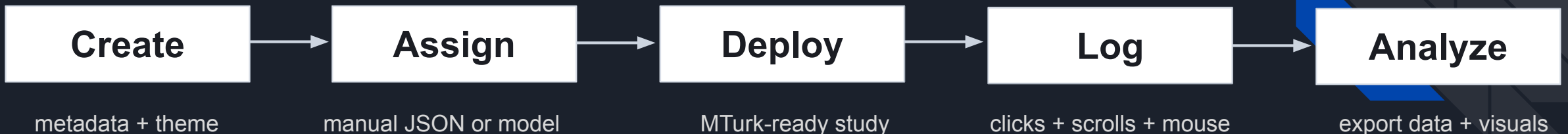
Existing experiments

View, modify or delete existing experiments using tables below.

Create a new Experiment

	HIT ID	Title	Replies	Active	Created at	
☐	3M9IY	Carousel Usability TN17	11	☒	1/5/2024	Logs Link Details Delete
☐	3X4X3	Carousel Usability TN16	25	☒	1/3/2024	Logs Link Details Delete
☐	CXLIF	Carousel Usability TP11	30	☐	12/3/2023	Logs Link Details Delete
☐	MSMJN	Carousel Usability TP10	9	☐	11/1/2023	Logs Link Details Delete

Reload



Modular Study Setup

CARE decomposes experiment creation into reusable configuration blocks.

CARE: A Carousel Evaluation Tool

Experiment Center New Experiment Settings Back

Create a new Experiment

Please complete the Experiment details below.

A Experiment Details

Experiment ID: MLQJP Created at: 2/8/2024 Experiment Description: Text or HTML

Experiment Name: Experiment Name Experiment Summary: Summary of activities

B Visual Options

Please select one of the following option

Basic Information Social Information Detailed Information

C Carousel Options

Please select the carousel options below:

Items per carousel: 5 Show Arrows Center Mode

Items to scroll: 5 Show Dots Daggable

D Surveys

Please insert the link to your pre and post task surveys:

Pre-task Survey: https://forms.gle/JdRfntzJ4UvU****

Post task Survey: <https://...>

E Task Assignment

Please provide information about the task in this experiment.

Automated Task Assignment

Recommendation Domain: Food and Recipes Embedding Model: a1-MiniLM-L6-v2 Difficulty Level: Intermediate

Manual Task Assignment

Task Name: Easy pasta recipe Task Summary: Pasta recipe with the fewest preparation steps

Task Description: Find and submit the pasta recipe with the fewest preparation steps.
 Note: Viewing recipe details might be required to confirm the step count.

Recommendation data: Choose File: food.json Validate and Upload

Cancel Save and Create

A Experiment metadata

C Carousel behavior options

E Task assignment and recommendations

B Visual theme selection

D External survey links

Proof of concept Experiment

Task

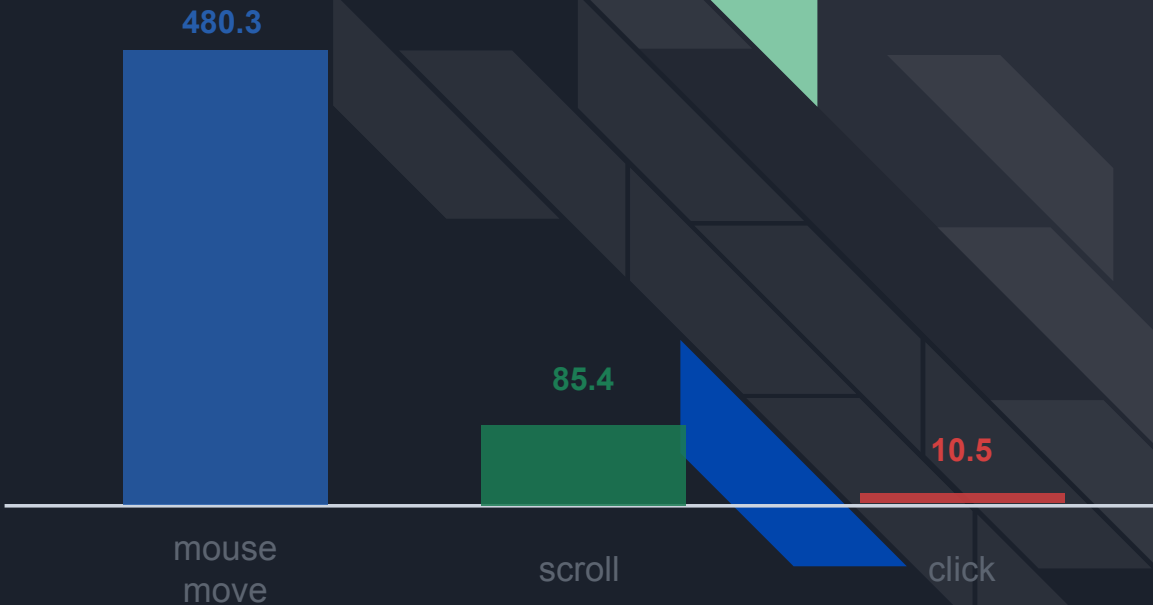
Find the lowest-rated sci-fi movie in a custom carousel of movie recommendations.

Logged metrics

Metric	Mean	Median	SD
Task Time (s)	26.53	23.21	12.31
Mouse Movement Distance (px)	2419.59	2002.20	1702.58
Mouse Speed (px/s)	184.80	176.92	93.57
Mouse Movement Events	480.3	400.8	215.1
Scroll Movement Events	85.40	62.00	63.64
Click Events	10.50	9.50	4.01

Interaction Metrics logged by the system. In this experiment, the "log inter seconds, px = pixels.

The output is not just a completion result. It is a fine-grained trace of how participants used the carousel.

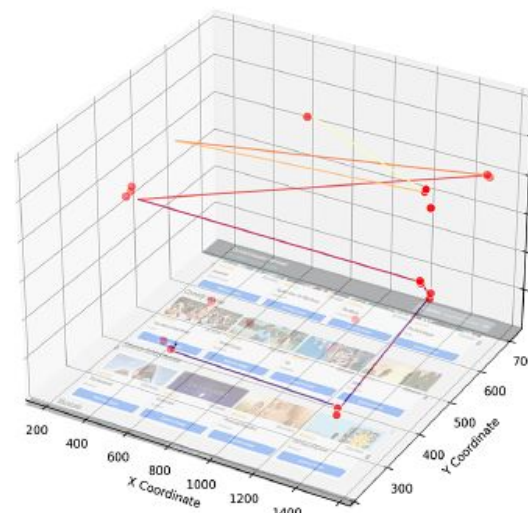


Possibilities and Limitations

Click density identifies where attention and action concentrate on the interface.



3D trajectories show the spatial and temporal



Limitations & Next Steps

- Best suited to short tasks because long sessions create large logs.
- Current studies are mostly predefined task-completion scenarios.
- Mouse movement is useful but imperfect as a proxy for attention.
- Planned extension: browser-based eye tracking with WebGazer.

RecGaze: The First Eye Tracking and User Interaction Dataset for Carousel Interfaces

Santiago de Leon-Martinez, Jingwei Kang, Robert Moro, Maarten de Rijke, Branislav Kveton, Harrie Oosterhuis, Maria Bielikova

Keywords: Carousel Interfaces, Browsing Behavior, Eye Tracking



Crime



Hostage (2005)

Genre: Action, Crime, Drama

Cast: Bruce Willis, Ben Foster, Jonathan Tucker

When a mafia accountant is taken hostage on his beat, a police officer – wracked by guilt from a prior stint as a negotiator – must negotiate the standoff, even as his own family is held captive by the mob.



Animation



Romance

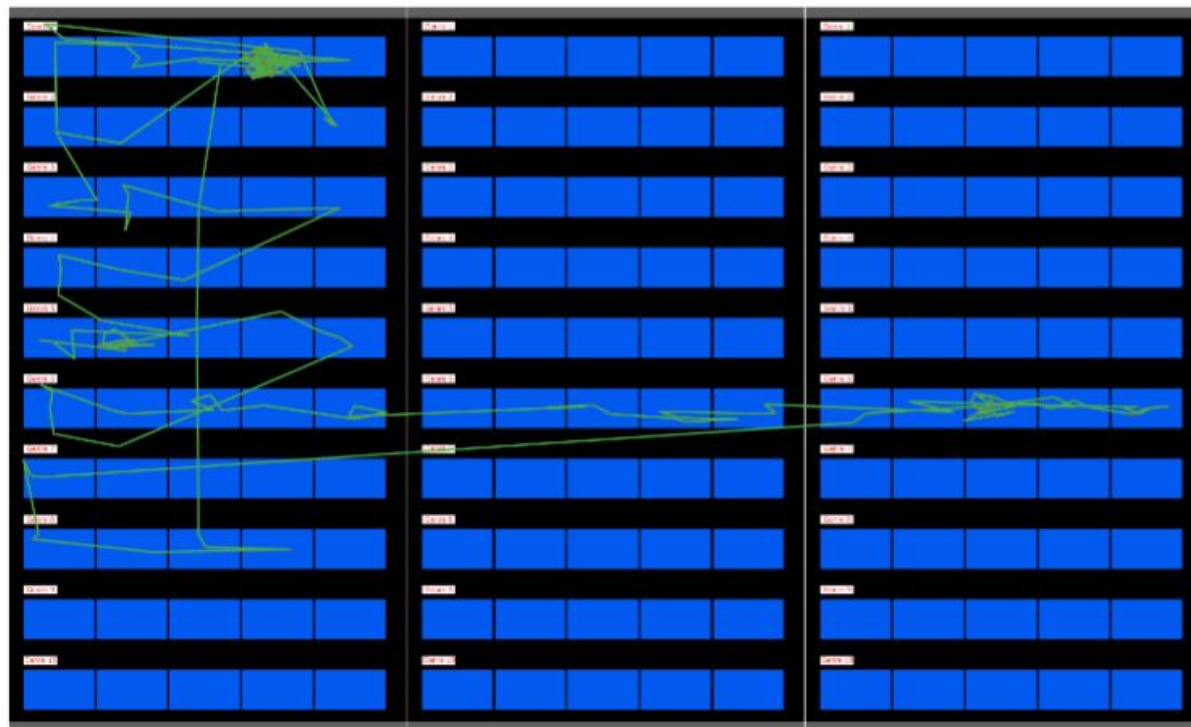


Table 1: Task summaries of interaction feedback, the number of movies selected (users can exit without selecting) and average screen time (until selection or exit).

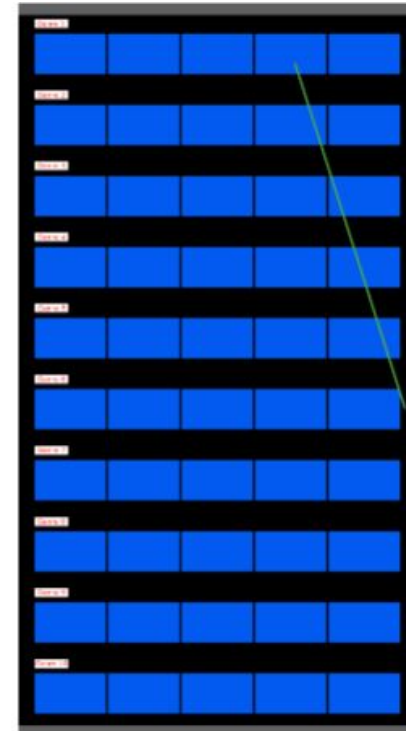
Task	Users	Screens	Interactions	Movie Selections	Avg. screen time
Free-browsing	87	30	2,607	2,432	56.99s
Semi-free	87	5	435	383	43.85s
Direct Search	87	5	435	424	11.57s
All Tasks	87	40	3,477	3,239	49.67s

The 1st comprehensive feedback dataset with gaze

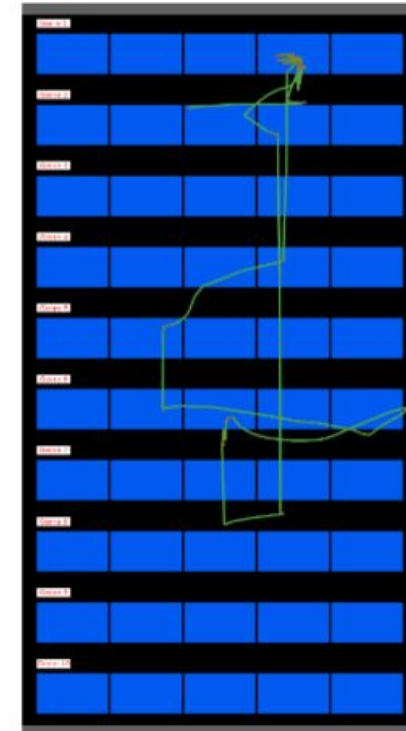
- Eye tracking (gaze)
- Clicks
- Cursor movements
- Selection Explanation
- More (user & item features, screen recordings, etc.)



(a) Fixation scan path with swipe differences



(b) Click scan path



(c) Cursor scan path

Figure 5: Visualizations of fixations, clicks and mouse cursor movements for one selected user and one free-browsing screen.

Table 2: Major features of the RecGaze Dataset. Features marked with * are part of the non-public dataset and square brackets [] refer to multiple responses. To be concise, we use shorthand names, combine similar features, and denote different columns using bold. For a detailed explanation of the features refer to the GitHub documentation.

Category	Feature name	Type	Example
User features	UserID	Multi	Location_ID
	Age*	Range	30-39
	Gender*	Text	Man
	Genre preferences	Multi	Q_Top: Action; Q_PREFERRED: [Action, Comedy]; Q_Ratings 1-5: [Action_5, Animation_1...]
	Interface expertise	Multi	3-4 times per week
	Movie expertise	Multi	1-2 times per month
	Decision-making	Int 1-7	Q_1: 1; Q_2: 3; Q_3: 5; Q_4: 7; Q_5: 4; Q_6: 5;
	Genre/Movie #	Text	Q_1: Yes, I felt overwhelmed by # of Genres; Q_2: No, I didn't feel overwhelmed by # of movies
Item features	MovieID	Int	372058 (ID from TMDB)
	Screen position	Multi	TaskID/Screen: 7; Carousel_position: 1; Genre: Horror; Movie_position: 6
	Title & Year	Multi	Title: Hostage; Year: 2005
	Description	Text	"When a mafia accountant is taken host..."
	Poster	Img	http://{url} (all downloaded jpegs are available on Zenodo)
Feedback & AOI label	Gaze (90 Hz)*	Multi	Time: 1.241s; Position: 120, 300 px;
	Fixation (90 Hz)*	Multi	Time: 1.241s; Position: 120, 300 px; AOI: Movie_372058; Duration: 0.891s
	Cursor (event)*	Multi	Time: 2.450s; Position: 100, 600 px; AOI: GenreText_1_Horror
	Click (event)*	Multi	Time: 10.903s; Position: 100, 142 px; AOI: BackSwipe_Genre_1_Horror
Summary feedback	Fixation sequence	Multi	Time: 1.241s; AOI: Movie_372058; Duration: 0.891s
	Cursor sequence	Multi	Time: 2.450s; AOI: GenreText_1_Horror; Duration: 1.902s
	Click sequence	Multi	Time: 10.903s; AOI: BackSwipe_Genre_1_Horror;
Selection explanation	Movie familiarity	Text	I have seen part of the movie
	Why selected?	Text	[Because of the poster, Because of the details, Other: "I like the director"]
Other	Screen recording*	Video	25 fps .avi video recording

Use Cases for the RecGaze Dataset

- Click models for carousel interfaces
- Offline evaluation
- User simulation
- Whole page optimization
- Impressions
- Gaze-based Recommenders (gaze as feedback)



Discussion & Q/A:

Discussion

Future Directions

Browsing Behavior

- User specific behaviors
- Biases
- Cursor proxy for ET

User Modeling

- Gaze-based systems
- Contextual user models
- Feedback besides clicks

Evaluation

- Duplicate items
- Inter-list interactions

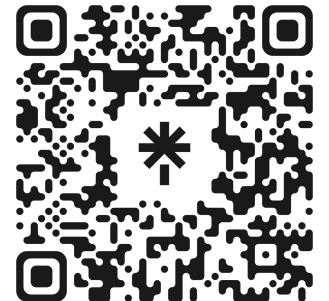
Other topics

- RecGaze Dataset
- CARE
- Mobile interfaces
- Social Media

Slides/Resources



Connect with
Santiago + Papers



We are hiring



<https://kinit.sk/careers/>

Web & User
Data
Processing



Green and
Secure
Environment



Natural
Language
Processing



Ethics and
Human Values
in Technology



Project
Management
& Marketing

